

23/1/25

சுருதல் புத்தகம்

Edl. B-132

பதிவுண்ணு

|| le & $\perp$ vector.

① condition for 2 vectors $\vec{A}$ & $\vec{B}$ to be perpendicular.

$$\vec{A} \cdot \vec{B} = 0 ;$$

$$\vec{A} \perp \vec{B}$$

② 2 vectors to be || le.

$$\vec{A} \times \vec{B} = 0 ; \vec{A} \parallel \vec{B}$$

problem:

① Show that given vectors $\vec{A} = 6\vec{a}_x + 2\vec{a}_y - 5\vec{a}_z$ and $\vec{B} = 5\vec{a}_x - 5\vec{a}_y + 4\vec{a}_z$ are perpendicular to each other.

sol:

2 vectors to be perpendicular:

$$\vec{A} \cdot \vec{B} = 0$$

Given:

$$\vec{A} \cdot \vec{B} = (A_x \cdot B_x) + (A_y \cdot B_y) + (A_z \cdot B_z)$$

$$= (6 \times 5) + (2 \times -5) + (-5 \times 4)$$

$$= 30 - 10 - 20 \Rightarrow 0$$

$$\vec{A} \cdot \vec{B} = 0$$

$\therefore \vec{A}$ & $\vec{B}$ vectors are perpendicular to each other.



2. show that the vector $\vec{A} = 4\vec{a}_x - 2\vec{a}_y + 2\vec{a}_z$ and $\vec{B} = -6\vec{a}_x + 3\vec{a}_y - 3\vec{a}_z$ are $\parallel$ to each other.

Sol:

Given:

$$\vec{A} =$$

$$\vec{B} =$$

2 vectors to be $\parallel$: $\vec{A} \times \vec{B} = 0$.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ 4 & -2 & 2 \\ -6 & 3 & -3 \end{vmatrix}$$

$$\Rightarrow \vec{a}_x [6 - 6] - \vec{a}_y [-12 + 12] + \vec{a}_z [12 - 12]$$

$$= \vec{a}_x (0) - \vec{a}_y (0) + \vec{a}_z (0)$$

$$\vec{A} \times \vec{B} = 0$$

$\therefore$ The given vector are $\parallel$ to each other.

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க. நதல் பத்தகர்

13/06/2001

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Solenoidal and irrotational vectors:

* Solenoidal vector:

$$\nabla \cdot \vec{A} = 0 \rightarrow \text{Divergence is Zero.}$$

* Irrotational vector:

$$\nabla \times \vec{A} = 0 \rightarrow \text{curl free vector}$$

[or]

conservative field.

Prob.

① show that the vector is solenoidal

$$\vec{H} = 3y^4 z^2 \vec{a}_x + 4x^3 z^2 \vec{a}_y + 3x^2 y^2 \vec{a}_z$$

condition for solenoidal. $\nabla \cdot \vec{H} = 0$ $z \rightarrow \text{component}$

$$\nabla \cdot \vec{A} = 0.$$

$$\nabla \cdot \vec{H} = \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z}$$

$$\nabla \cdot \vec{H} = \frac{\partial}{\partial x} [3y^4 z^2] + \frac{\partial}{\partial y} [4x^3 z^2] + \frac{\partial}{\partial z} [3x^2 y^2]$$

$\Rightarrow$

$$0 + 0 + 0.$$

no y term

$$\nabla \cdot \vec{H} = 0 \rightarrow \text{The given vector is Zero.}$$

Q. Find the constant 'c' such that the Vector.
 $\vec{A} = (x+ay) \vec{i} + \frac{(y+bz)}{Ay} \vec{j} + \frac{(x+Cz)}{Az} \vec{k}$ will be solenoidal.

Sol:

Condition for solenoidal $\nabla \cdot \vec{A} = 0$ (divergence of)

$$\nabla \cdot \vec{A} = \frac{\partial Ax}{\partial x} + \frac{\partial Ay}{\partial y} + \frac{\partial Az}{\partial z}$$

$$= \frac{\partial}{\partial x} [x+ay] + \frac{\partial}{\partial y} [y+bz] + \frac{\partial}{\partial z} [x+Cz]$$

$$0 = [1+0] + [1+0] + [0+C]$$

$$1+1+C = 0$$

$$2+C = 0$$

$$\boxed{C = -2}$$

Q. Show that $\vec{H} = x^2 \vec{ax} + y^2 \vec{ay} + z^2 \vec{az}$ is a conservative field.

Sol:

condition for conservative field:

$$\nabla \times \vec{H} = 0$$

$$\nabla \times \vec{H} = \begin{vmatrix} \vec{ax} & \vec{ay} & \vec{az} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$\downarrow$
 ax

$\downarrow$
 ay

$\downarrow$
 az

(component)

$$\nabla \cdot \nabla \times \vec{H} = \frac{\partial}{\partial x} \left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right] - \frac{\partial}{\partial y} \left[\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z} \right] + \frac{\partial}{\partial z} \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right]$$

$$\Rightarrow \frac{\partial^2 H_z}{\partial x \partial y} - \frac{\partial^2 H_y}{\partial x \partial z} - \frac{\partial^2 H_z}{\partial y \partial x} + \frac{\partial^2 H_x}{\partial y \partial z} + \frac{\partial^2 H_y}{\partial x \partial z} - \frac{\partial^2 H_x}{\partial y \partial z}$$

$$\boxed{\nabla \cdot \nabla \times \vec{H} = 0}$$

Hence the given Null identity is proved.

② prove that.

$$\nabla \times \nabla v = 0$$

[Doesn't have any direction.]

⇒ Question ⇒ (v or F) or any scalar value.
curl of gradient of a scalar is zero.

sol:

v → Any scalar field.

→ Gradient of a scalar:

$$\nabla v = \frac{\partial v}{\partial x} \vec{a}_x + \frac{\partial v}{\partial y} \vec{a}_y + \frac{\partial v}{\partial z} \vec{a}_z$$

x, y, z
↓
component

→ curl of ∇v .

$$\nabla \times \nabla v =$$

$$\begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \end{vmatrix}$$

Null Identities.

i) $\frac{\partial}{\partial t} \nabla \cdot \nabla \times \vec{H} = 0$ ~~✗~~

The divergence of curl of vector is equal to zero

ii) $\nabla \times \nabla \phi = 0$

The curl of the gradient of any scalar is equal to zero.

Prove that $\nabla \cdot \nabla \times \vec{H} = 0$

Question may be. $\vec{A}, \vec{F}$ (or) the divergence of curl of any vector is equal to zero.

Sol:

Let $\vec{H} = H_x \vec{a}_x + H_y \vec{a}_y + H_z \vec{a}_z$ → unit vectors in the particular direction.

$$\nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

$$\nabla \times \vec{H} \Rightarrow \vec{a}_x \left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right] - \vec{a}_y \left[\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z} \right] + \vec{a}_z \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right]$$

with this (-) sign. $\vec{a}_y$

$$\nabla \cdot \frac{\nabla \times \vec{H}}{A} \Rightarrow \nabla \cdot \vec{A} =$$

Divergence of a vector:

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

ii) condition for irrotational.

$$\nabla \times \vec{E} = 0$$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix}$$

$$= \vec{a}_x \left[\frac{\partial xy}{\partial y} - \frac{\partial xz}{\partial z} \right] - \vec{a}_y \left[\frac{\partial xy}{\partial x} - \frac{\partial yz}{\partial z} \right]$$

$$+ \vec{a}_z \left[\frac{\partial xz}{\partial x} - \frac{\partial yz}{\partial y} \right]$$

$$= \vec{a}_x [x - x] - \vec{a}_y [y - y] + \vec{a}_z [z - z]$$

$$\boxed{\nabla \times \vec{E} = 0}$$

∴ Hence the given vector $\vec{E}$ is

solenoidal & irrotational.

$$\nabla \times \vec{H} = \vec{a}_x \left[\frac{\partial z^2}{\partial y} - \frac{\partial y^2}{\partial z} \right] - \vec{a}_y \left[\frac{\partial z^2}{\partial x} - \frac{\partial x^2}{\partial z} \right] +$$

$$\vec{a}_z \left[\frac{\partial y^2}{\partial x} - \frac{\partial x^2}{\partial y} \right]$$

$$= \vec{a}_x [0] - \vec{a}_y [0] + \vec{a}_z [0]$$

$$\nabla \times \vec{H} = 0$$

Hence the given vector is a conservative vector.

4) verify whether the given vector is both solenoidal & irrotational.

Sol:

$$\vec{E} = yz \vec{a}_x + xz \vec{a}_y + xy \vec{a}_z$$

i) condition for solenoidal.

$$\nabla \cdot \vec{E} = 0$$

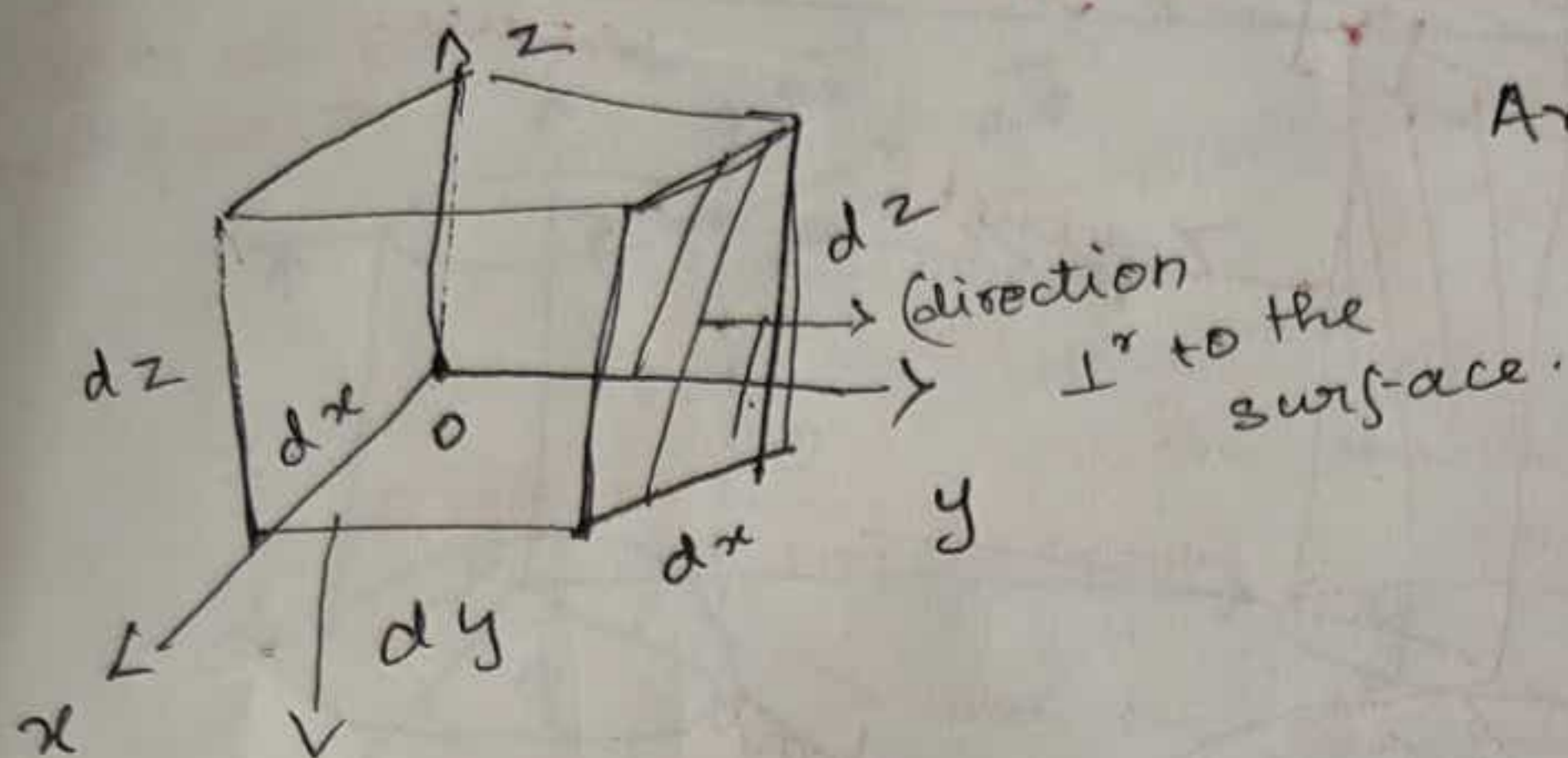
$$\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \cdot \vec{E} = \frac{\partial}{\partial x} [yz] + \frac{\partial}{\partial y} [xz] + \frac{\partial}{\partial z} [xy]$$

$$\nabla \cdot \vec{E} = 0$$

$$|dl| = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$$

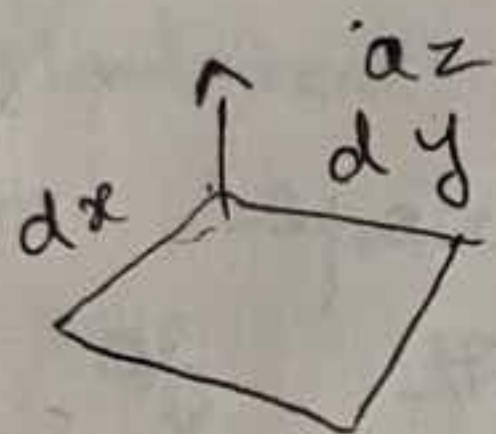
The differential area.



$$\text{Area} = l \times b.$$

$$\vec{ds}_y = dx dz \vec{a}_y$$

$$\vec{ds}_x = dy dz \vec{a}_x$$



$$\vec{ds}_z = dx dy \vec{a}_z$$

differential volume:

$$dv = dx dy dz$$

where:

x, y & z are the component vectors.

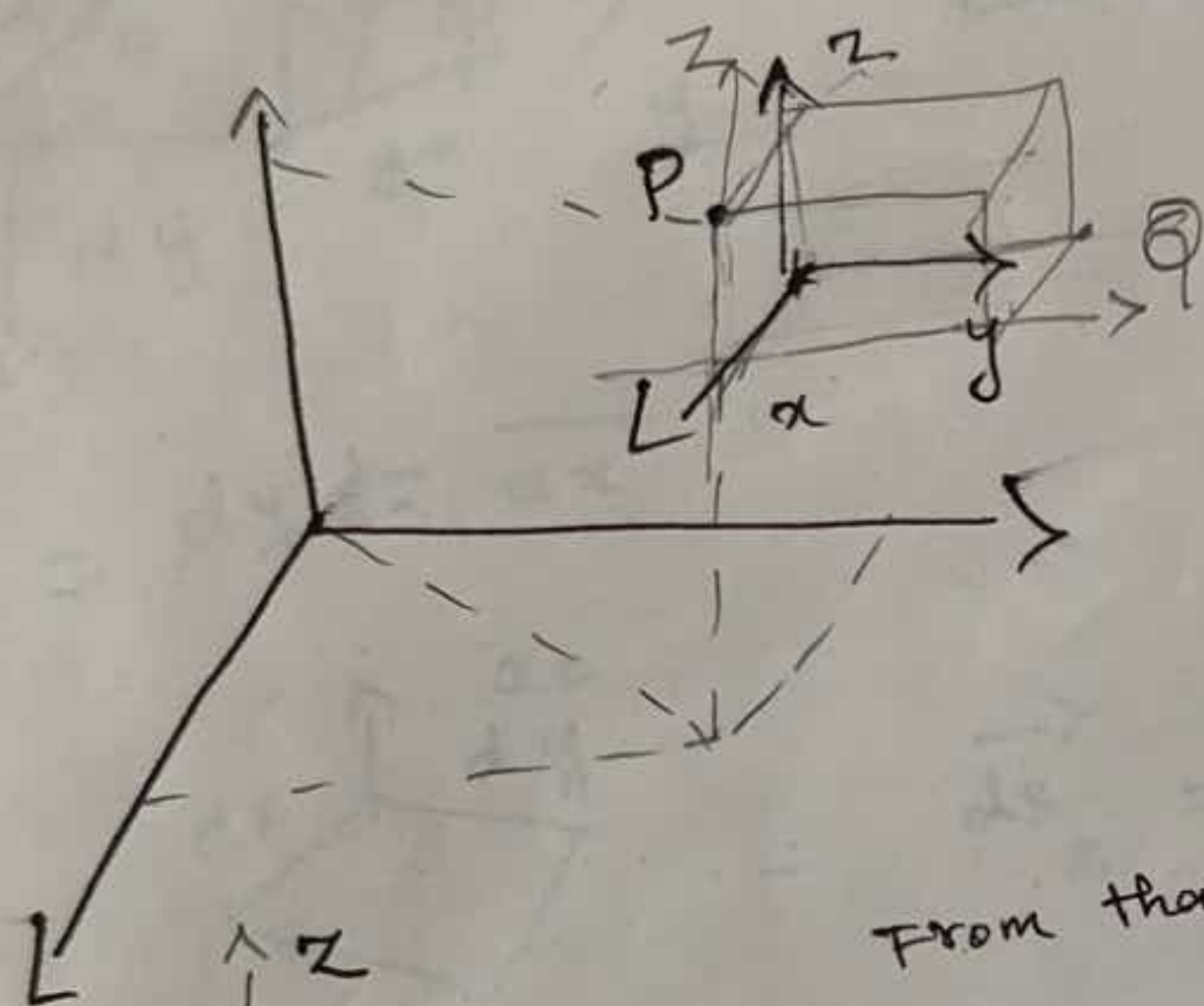
$\vec{a}_x, \vec{a}_y$ & $\vec{a}_z$ are unit vectors.

if it is a points represent distance vector.

[Differential length, surface, volume]

Increase a length of this point

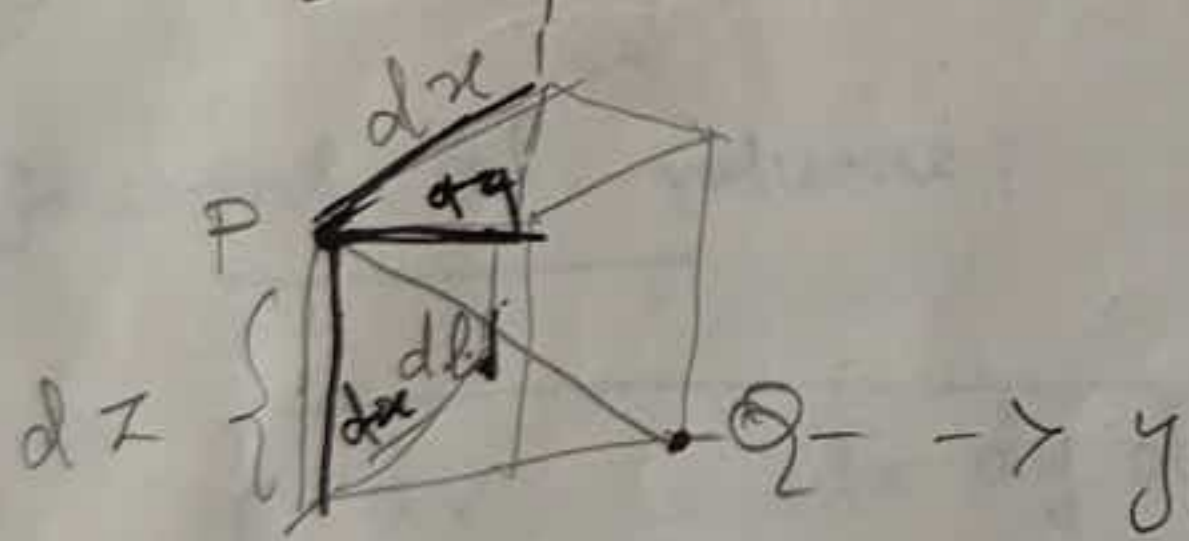
$\Rightarrow$



① Draw (↑ the length of P by point (dx, dy, dz))

$P \rightarrow x, y, z$.

From that point $\rightarrow$ by $x+dx, y+dy, z+dz$



$Q (x+dx, y+dy, z+dz)$.

$$dl = (x+dx, y+dy, z+dz) - (x, y, z)$$

$$d\vec{l} \Rightarrow dx \vec{a}_x + dy \vec{a}_y + dz \vec{a}_z$$

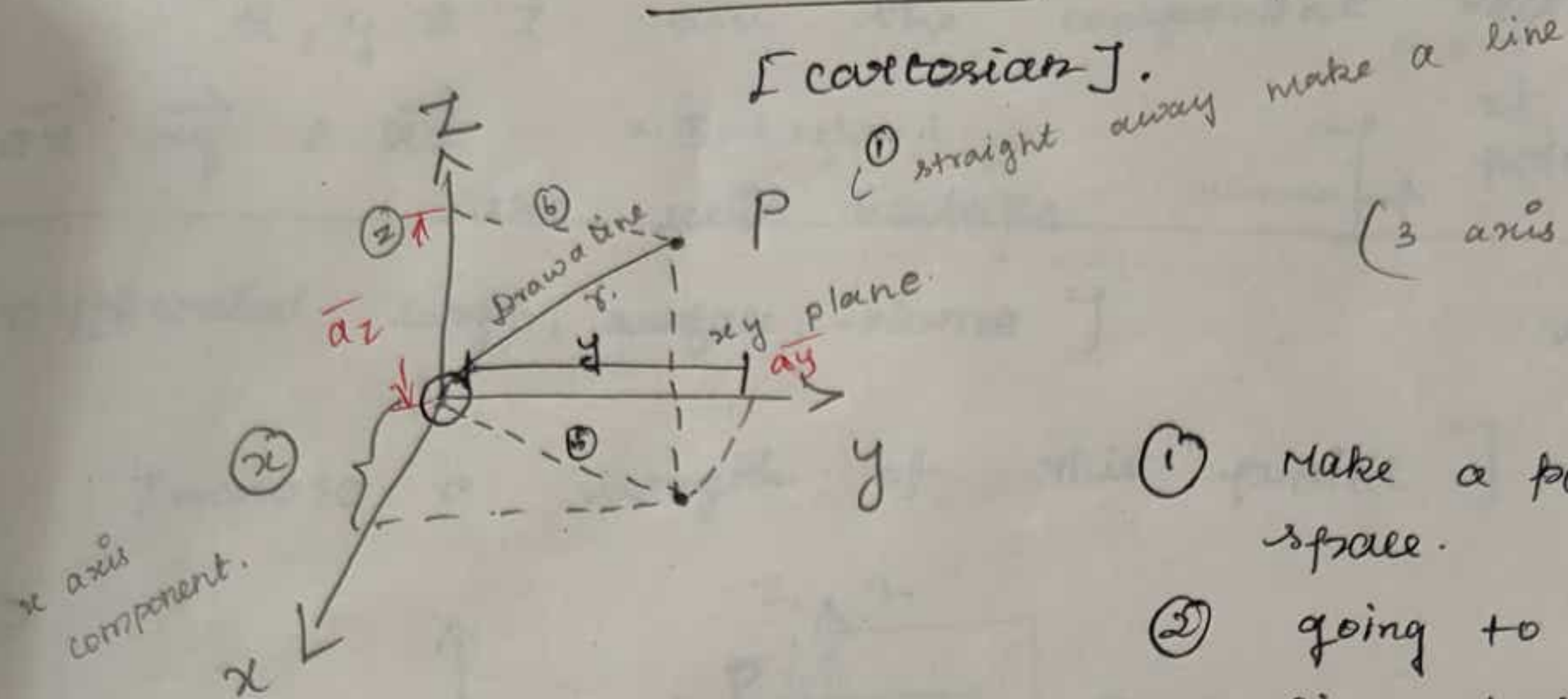
$\Rightarrow$ The differential length is nothing but diagonal value of Q.



$$dl = \text{End point} - \text{Initial point}.$$

Rectangular co-ordinate system.

[Cartesian].



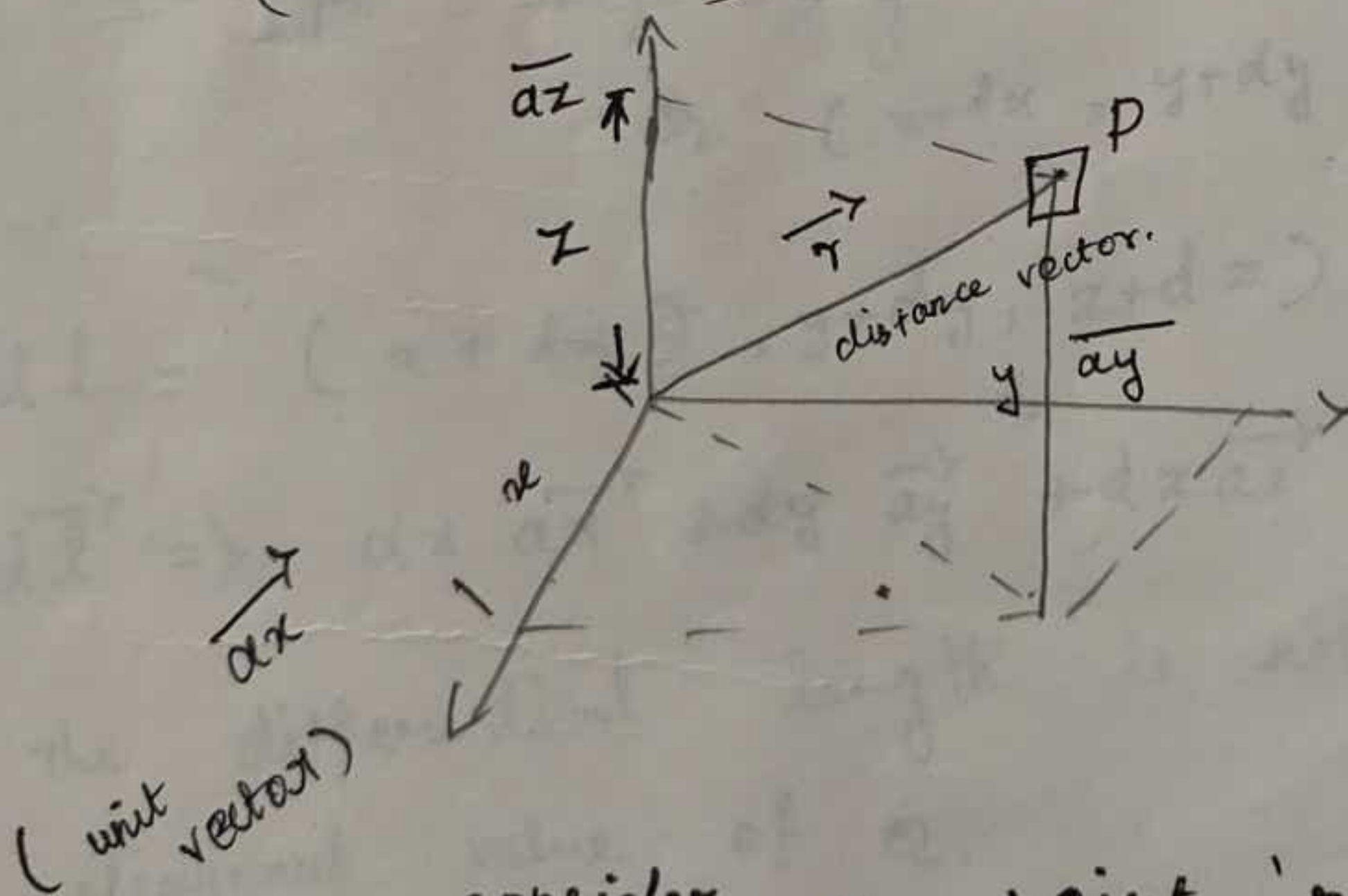
(3 axis $\perp$ to each other)

x & y [x axis component for point]

⑤ To get z axis value. for that make a line from origin to P.

⑥ P $\rightarrow$ Distance vector. (draw one line in xy.

⑥ same line drop at the z axis. (5 6 6 // le.)



direct part of y is $\overline{ay}$ & x is $\overline{ax}$.

consider a point 'P' in space at a distance 'r' from the Origin.

$$\vec{r} = x \vec{ax} + y \vec{ay} + z \vec{az}$$

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$$= \bar{a}_x \left[\frac{\partial^2 v}{\partial y \partial z} - \frac{\partial^2 v}{\partial y \partial z} \right] - \bar{a}_y \left[\frac{\partial^2 v}{\partial x \partial z} - \frac{\partial^2 v}{\partial x \partial z} \right] +$$

$$\bar{a}_z \left[\frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial x \partial y} \right]$$

$$= 0$$

$$\boxed{\nabla \times \nabla v = 0} \text{ hence proved //}$$

other. The Co-ordinate Systems.

↓ 3 types.

i) cartesian (or) Rectangular (x, y, z)

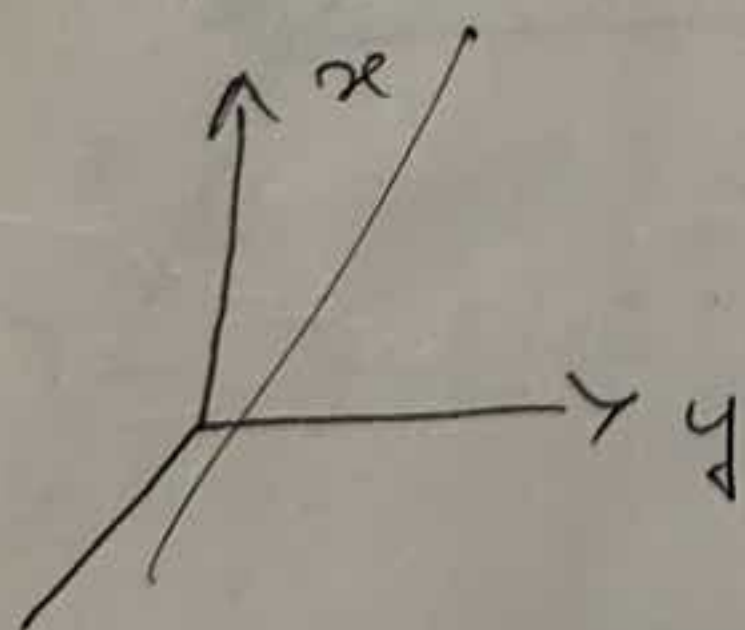
ii) cylindrical c.s (ρ, ϕ, z)

iii) spherical c.s (r, θ, ϕ)

Def:

cartesian co-ordinate systems:

* co ordinate system is a system which is used to represent a point in space.



→ It is used to describe a vector accurately with specific length, directions, angles, projections and components.

[Any point is define by intersecting 3 mutually $\perp$ planes.]



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$r \rightarrow$ distance from the origin to point 'p' (or)
radius of sphere; $0 \leq r \leq \infty$

$\theta \rightarrow$ angle between z -axis and the radius 'r'

$$0 \leq \theta \leq \pi$$

$\phi \rightarrow$ angle made by point 'p' w.r. to x axis.

$$0 \leq \phi < 2\pi$$

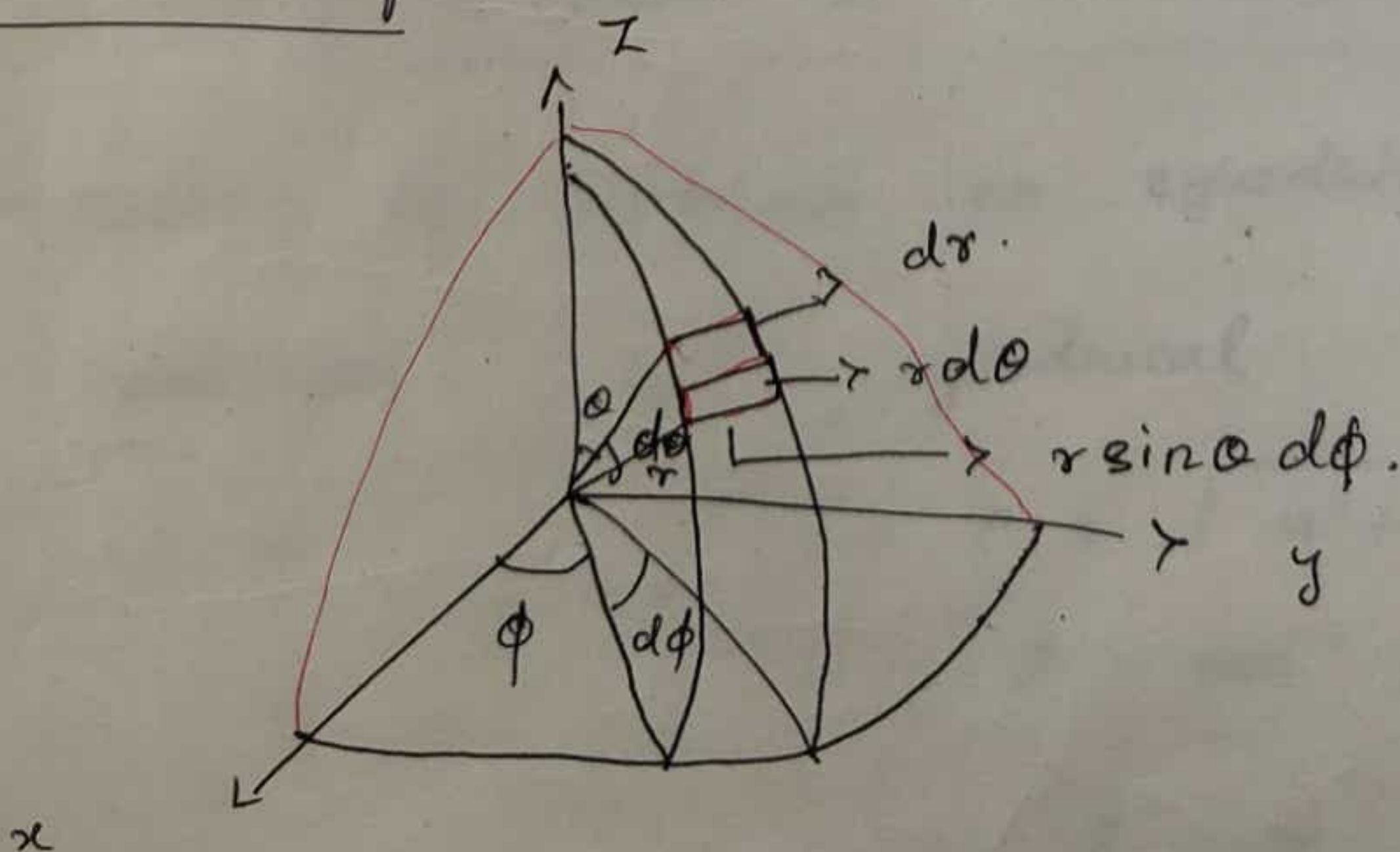
* In spherical co-ordinates system, any point 'p' in space is considered as the intersection of three surfaces.

$r = \text{constant}$: spherical surface.

$\theta = \text{constant}$: conical surface.

$\phi = \text{constant}$: plane surface.

differential length:



of the conductor is directly charge, it contains

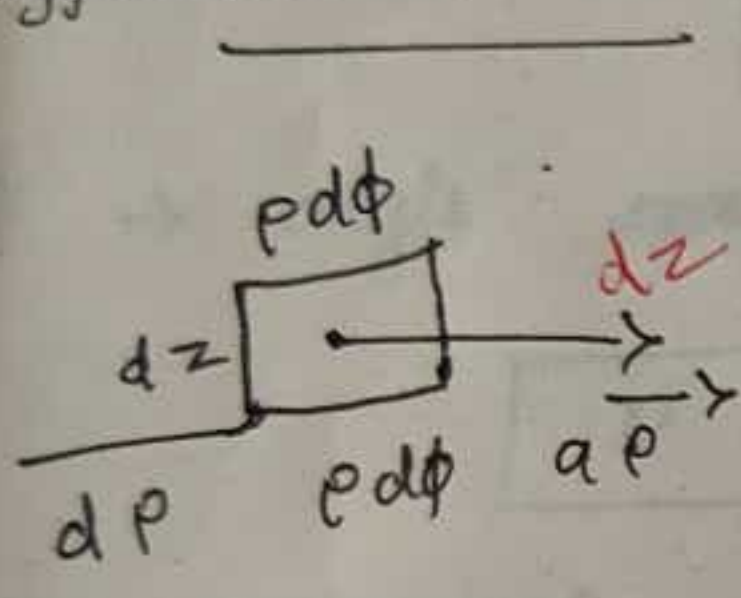
for sphere
 $\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial V}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial V}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = -\rho / \epsilon_0$

$$d\vec{l} = [(r + dr), (\phi + r d\phi), (z + dz)] - [r, \phi, z]$$

$$d\vec{l} = dr \vec{a}_r + r d\phi \vec{a}_\phi + dz \vec{a}_z$$

$$|dl| = \sqrt{(dr)^2 + (r d\phi)^2 + (dz)^2}$$

Differential surface:



$$dS_r = r d\phi dz \vec{a}_r$$

$$dS_\phi = dr dz \vec{a}_\phi$$

$$dS_z = r d\phi dr \vec{a}_z$$

Volume: [Differential]

$$dv = r dr d\phi dz$$

Spherical Coordinate system

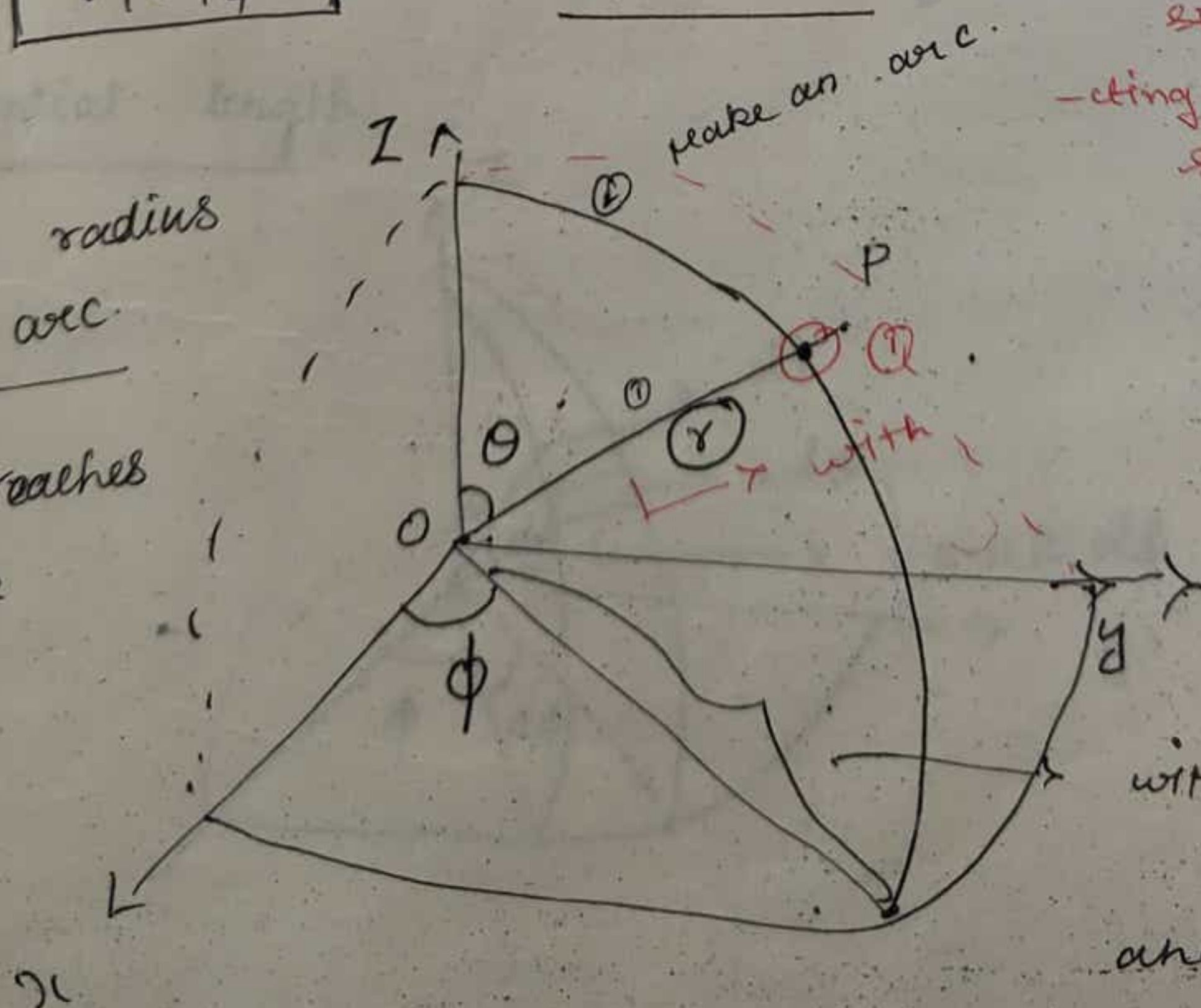
$$r, \theta, \phi$$

3 Co-ordinates

① point in a space by intersecting mutually $\perp$ surfaces

① with this radius make an arc

② circle reaches x,y plane



with this radius draw another circle

of the conductor is directly charge. it contain.

for sphere

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

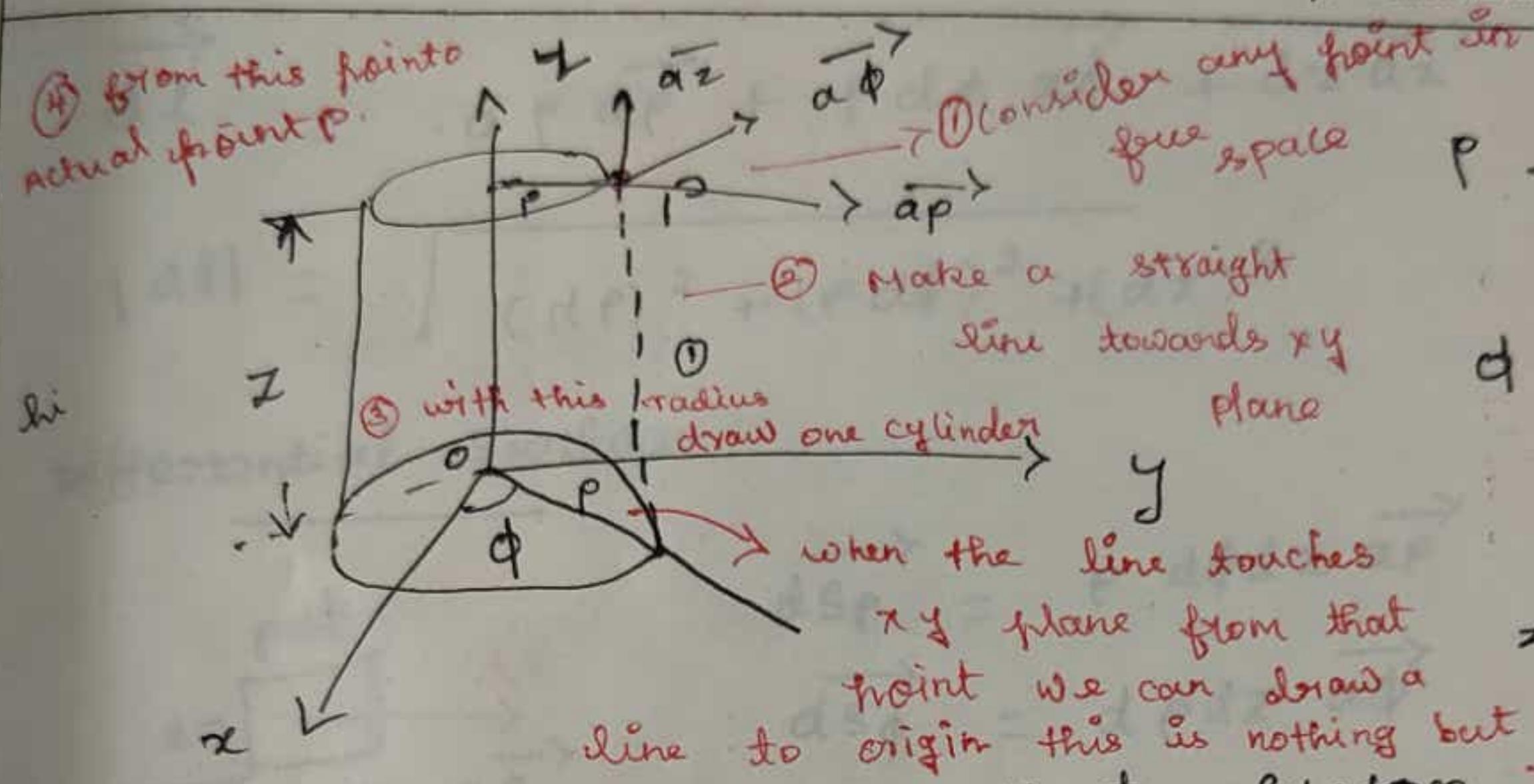
Cylindrical Coordinate System.

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$[\rho, \phi, z]$

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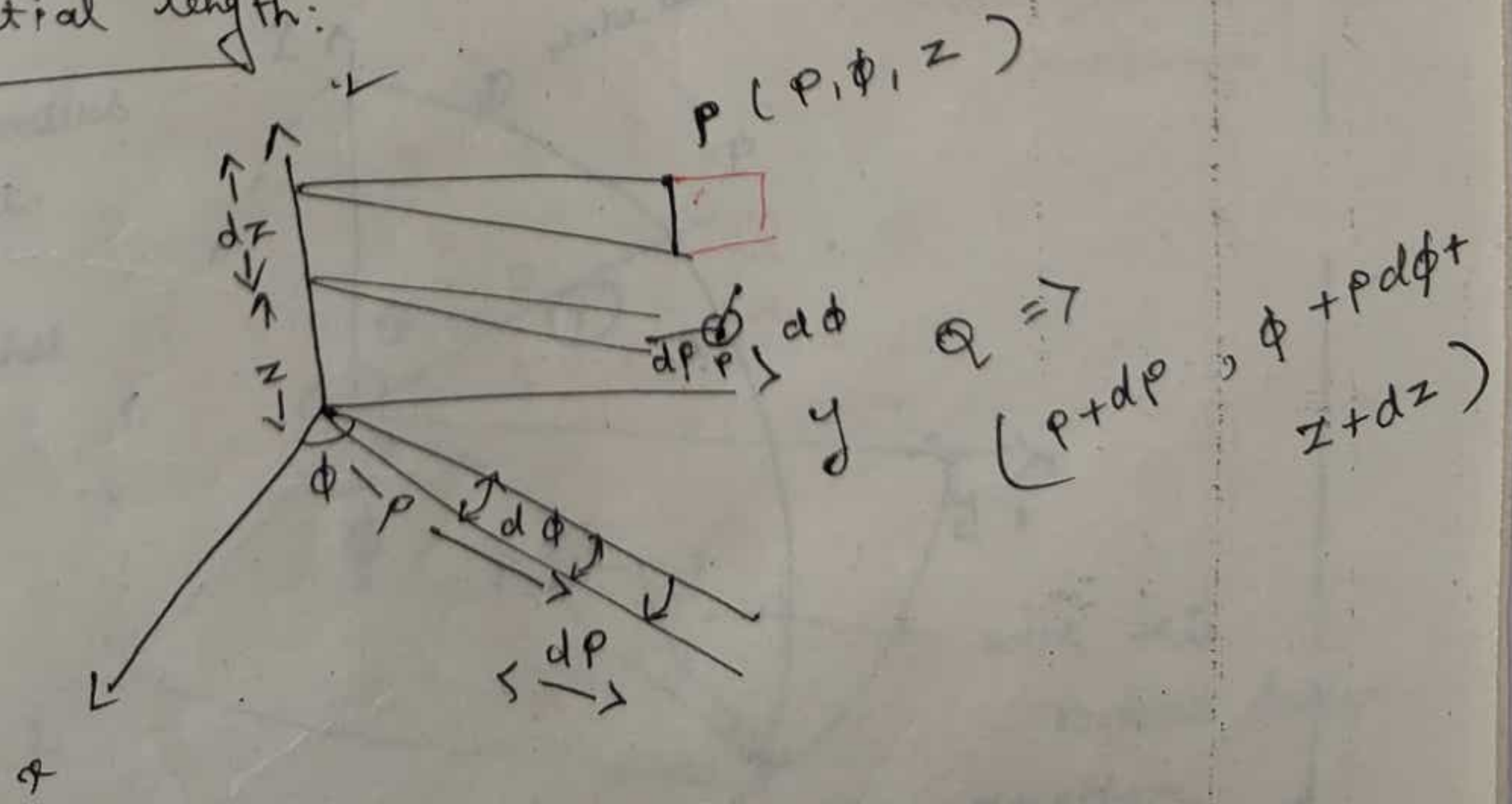


$\rho \rightarrow$ radius of the cylinder.
 $\phi \rightarrow$ angle made by ' ρ ' w.r. to ' x '
 $z \rightarrow$ height of the cylinder.

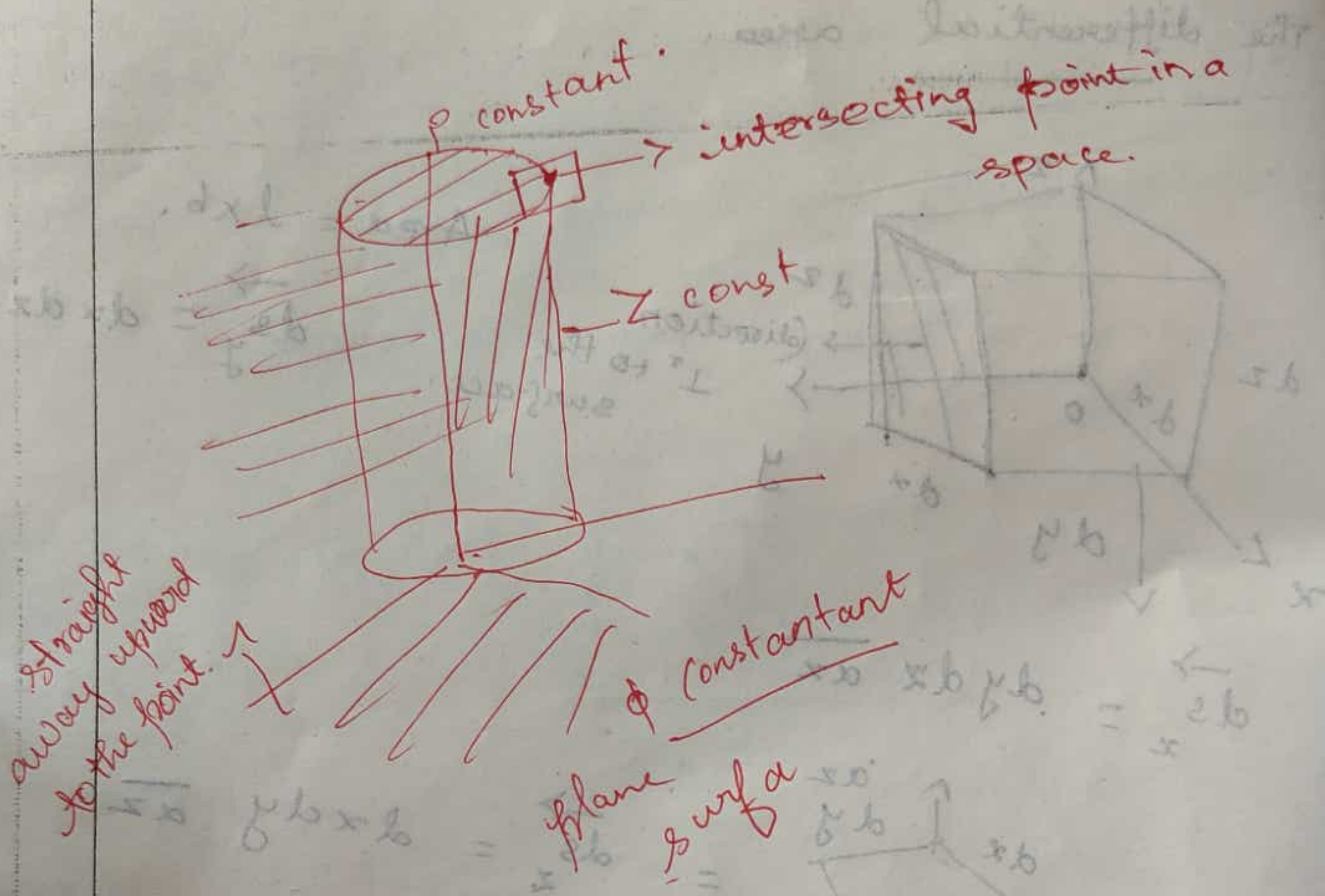
In cylindrical Co-ordinate System, any point in space is considered as the intersection of three mutually perpendicular surfaces.

- $\rho = \text{constant}$; create circular cylinder surface.
- $\phi = \text{constant}$; a plane surface.
- $z = \text{constant}$; a plane.

* differential length:



$$\sqrt{(x-b)^2 + (y-b)^2 + (z-b)^2} = |2b|$$



different volumes

$$\sqrt{(x-b)^2 + (y-b)^2 + (z-b)^2} = |2b|$$

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② Transform the given cylindrical co-ordinates into cartesian co-ordinates.

$$\rho = 1$$

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$$\phi = 45^\circ$$

$$z = 2$$

sol:

cylindrical : $\rho = 1$; $\phi = 45^\circ$, $z = 2$

cartesian : $x = \rho \cos \phi = (1) \cos 45^\circ = 0.707$

$$y = \rho \sin \phi = (1) \sin (45^\circ) = 0.707$$

$$z = 2$$

$$\therefore x = 0.707, y = 0.707, z = 2 //$$

Ex: Express the given vector in cylindrical co-ordinate.

$$\vec{A} = 4\vec{a}_x + 2\vec{a}_y + 3\vec{a}_z$$

[Rectangular vec to cylindrical.]

sol:

$$\text{cylindrical : } \vec{A} = A_\rho \vec{a}_\rho + A_\phi \vec{a}_\phi + A_z \vec{a}_z$$

$$A_\rho = A_x \cos \phi + A_y \sin \phi \quad \downarrow \text{unit vector.}$$

$$= 4 \cos 26.56^\circ + 2 \sin 26.56^\circ$$

$$= 4 \times 0.8945 + 2 \times 0.447$$

$$A_\rho = 4.47$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$= -4 \sin (26.56^\circ) + 2 \cos (26.56^\circ)$$

$$= -4 \times 0.447 + 2 \times 0.894$$

$$A_\phi = 0$$

Given:

$$A_x = 4$$

$$A_y = 2$$

$$A_z = 3$$

$$\phi = \tan^{-1} (y/x)$$

$$= \tan^{-1} (2/4)$$

$$\phi = 26.56^\circ$$

$$\boxed{A_z = 3}$$

total of the conductor is directly
the charge. it contains

Conversion of cartesian to cylindrical
co-ordinate system.

Important formula:

cartesian

x

y

z

cylindrical.

$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}(y/x)$$

$$z = z$$

$\Rightarrow$ cylindrical to cartesian.

Prob.

1. Transform the cartesian co-ordinates. $(2, 1, 3)$
into cylindrical co-ordinates.

sol:

cartesian ; $x = 2$; $y = 1$, $z = 3$.

cylindrical co-ordinates.

$$\rho = \sqrt{x^2 + y^2} = \sqrt{2^2 + 1^2} = \sqrt{5} = 2.236$$

$$\phi = \tan^{-1}(y/x) = \tan^{-1}(1/2) = 26.56^\circ$$

$$z = 3.$$

Ans :

cylindrical co-ordinates :

$$(2.236, 26.56^\circ, 3) //$$

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ii) conversion of cartesian to spherical:

cartesian $\rightarrow$ spherical. x

$$r = \sqrt{x^2 + y^2 + z^2}$$

 y

$$\theta = \cos^{-1}(z/r)$$

 z

$$\phi = \tan^{-1}(y/x)$$

iii) conversion of cylindrical to cartesian.

 cylindrical $\rightarrow$ cartesian. ρ

$$x = \rho \cos \phi$$

 ϕ

$$y = \rho \sin \phi$$

 z

$$z = z.$$

iv) conversion of spherical to cartesian:

spherical $\rightarrow$ cartesian. r

$$x = r \sin \theta \cdot \cos \phi$$

 θ

$$y = r \sin \theta \cdot \sin \phi$$

 ϕ

$$z = r \cos \theta.$$

differential length:

$$d\vec{l} = dr\vec{a}_r + r d\theta \vec{a}_\theta + r \sin\theta d\phi \vec{a}_\phi$$

$$|dl| = \sqrt{(dr)^2 + (r d\theta)^2 + (r \sin\theta d\phi)^2}$$

Magnitude.

The differential surface:

$$d\vec{s} = dr (r d\theta) \vec{a}_\phi$$

$$\Rightarrow r d\theta dr \vec{a}_\phi$$

$$d\vec{s} = (r d\theta) (r \sin\theta d\phi) \vec{a}_r$$

$$= r^2 \sin\theta d\theta d\phi \vec{a}_r$$

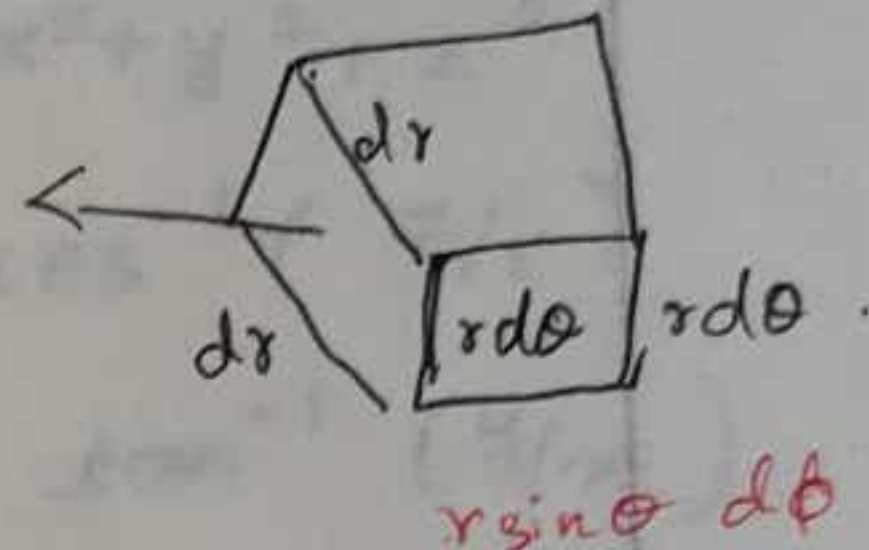
$$d\vec{s} = r \sin\theta d\phi dr \vec{a}_\theta$$

The differential volume:

$$dv = dr (r d\theta) (r \sin\theta d\phi)$$

$$dv = r^2 \sin\theta dr d\theta d\phi$$

(x, y, z sides)



Conversion of co-ordinate systems.

i) conversion of cartesian to cylindrical:

cartesian $\rightarrow$ cylindrical

x

$$\rho = \sqrt{x^2 + y^2}$$

y

$$\phi = \tan^{-1} (y/x)$$

z

$$z = z$$

potential of the conductor is charge, it contain.

$Hx \rightarrow$ common
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$$\nabla^2 H_x = \frac{\partial^2 H_x}{\partial x^2} + \frac{\partial^2 H_x}{\partial x \partial y} + \frac{\partial^2 H_x}{\partial x \partial z} - \frac{\partial^2 H_x}{\partial x^2} +$$

$$\frac{\partial^2 H_x}{\partial y^2} + \frac{\partial^2 H_x}{\partial z^2}$$

$$+ \frac{\partial^2 H_y}{\partial y^2} + \frac{\partial^2 H_x}{\partial x \partial y} + \frac{\partial^2 H_z}{\partial y \partial z} - \frac{\partial^2 H_y}{\partial x^2} + \frac{\partial^2 H_y}{\partial y^2} +$$

$$\frac{\partial^2 H_y}{\partial z^2}$$

$$+ \frac{\partial^2 H_z}{\partial z^2} + \frac{\partial^2 H_x}{\partial x \partial z} + \frac{\partial^2 H_y}{\partial y \partial z} - \frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} +$$

$$\frac{\partial^2 H_z}{\partial z^2}$$

$$\Rightarrow \frac{\partial}{\partial x} \left[\frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} \right]$$

$$\frac{\partial}{\partial y} \text{ common}$$

$$\frac{\partial}{\partial z}$$

$$\nabla^2 H_z = \frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + \frac{\partial^2 H_z}{\partial z^2}$$

∇^2

The Laplacian Operator:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\sigma \times \nabla \times \vec{H} = \vec{a}_x \left\{ \frac{\partial^2 H_y}{\partial x \partial y} - \frac{\partial^2 H_x}{\partial y \partial z} - \frac{\partial^2 H_x}{\partial z^2} + \frac{\partial^2 H_z}{\partial x \partial z} \right\}$$

$\frac{\partial^2 H_x}{\partial x^2} \rightarrow \text{missing}$

$$+ \vec{a}_y \left\{ - \frac{\partial^2 H_y}{\partial x^2} + \frac{\partial^2 H_x}{\partial x \partial y} + \frac{\partial^2 H_z}{\partial y \partial z} - \frac{\partial^2 H_y}{\partial z^2} \right\}$$

$$+ \vec{a}_z \left\{ \frac{\partial^2 H_x}{\partial x \partial z} - \frac{\partial^2 H_z}{\partial x^2} - \frac{\partial^2 H_z}{\partial y^2} + \frac{\partial^2 H_y}{\partial y \partial z} \right\}$$

$$- \frac{\partial^2 H_y}{\partial y^2} \rightarrow \text{missing}$$

$$- \frac{\partial^2 H_z}{\partial z^2}$$

$$\Rightarrow - \frac{\partial^2 H_x}{\partial x^2} + \frac{\partial^2 H_x}{\partial x^2}$$

$$- \frac{\partial^2 H_y}{\partial y^2} + \frac{\partial^2 H_y}{\partial y^2}$$

$$- \frac{\partial^2 H_z}{\partial z^2} + \frac{\partial^2 H_z}{\partial z^2}$$

$$= \vec{a}_x \left\{ \frac{\partial^2 H_y}{\partial x \partial y} + \frac{\partial^2 H_z}{\partial x \partial z} - \frac{\partial^2 H_x}{\partial y^2} - \frac{\partial^2 H_x}{\partial z^2} - \frac{\partial^2 H_x}{\partial x^2} + \frac{\partial^2 H_x}{\partial x^2} \right\}$$

$$+ \vec{a}_y \left\{ \frac{\partial^2 H_x}{\partial x \partial y} + \frac{\partial^2 H_z}{\partial y \partial z} - \frac{\partial^2 H_y}{\partial x^2} - \frac{\partial^2 H_y}{\partial z^2} - \frac{\partial^2 H_y}{\partial y^2} + \frac{\partial^2 H_y}{\partial y^2} \right\}$$

$$+ \vec{a}_z \left\{ \frac{\partial^2 H_x}{\partial x \partial z} + \frac{\partial^2 H_y}{\partial y \partial z} - \frac{\partial^2 H_z}{\partial x^2} - \frac{\partial^2 H_z}{\partial y^2} - \frac{\partial^2 H_z}{\partial z^2} + \frac{\partial^2 H_z}{\partial z^2} \right\}$$

real of the conductor is directly charge. it contain.

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

* Prove the Identity:

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$$\nabla \times \nabla \times \vec{H} = \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H}$$

Proof

Let $\vec{H} = H_x \vec{a}_x + H_y \vec{a}_y + H_z \vec{a}_z$.

L.H.S

Taking curl of $\vec{H}$

$$\nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

$$\nabla \times \vec{H} = \vec{a}_x \left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right] - \vec{a}_y \left[\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z} \right] + \vec{a}_z \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right]$$

Taking curl on both sides.

$$\nabla \times \nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right] & \left[\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right] & \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right] \end{vmatrix}$$

$$\begin{aligned} \nabla \times \nabla \times \vec{H} &= \vec{a}_x \left\{ \frac{\partial}{\partial y} \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \right\} \\ &\quad - \vec{a}_y \left\{ \frac{\partial}{\partial x} \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \right\} \\ &\quad + \vec{a}_z \left\{ \frac{\partial}{\partial x} \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \right\} \end{aligned}$$

val of the conductor is directly charge. it contain

Ans:-

$$\vec{A} = A\rho \vec{a}_\rho + A\phi \vec{a}_\phi + Az \vec{a}_z$$

$$\Rightarrow 47 \vec{a}_\rho + 3 \vec{a}_z //$$

Cartesian to spherical coordinates

spherical to Cartesian.

Problem:

1. Transform the Cartesian coordinates to Spherical co-ordinates:

$$x=2, y=1, z=3$$

Sol:

$$r, \theta, \phi$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{2^2 + 1^2 + 3^2} = \sqrt{14} \quad r = 3.74$$

$$\theta = \cos^{-1}(z/r) = \cos^{-1}(3/3.74) \quad \therefore \theta = 36.66^\circ$$

$$\phi = \tan^{-1}(y/x) = \tan^{-1}(1/2) \quad \therefore \phi = 26.56^\circ$$

Ans:

2. Find the Cartesian co-ordinates.

$$r=4, \theta=25^\circ, \phi=120^\circ$$

Sol:

$$r, \theta, \phi \Rightarrow (x, y, z)$$

$$y = r \sin \theta \sin \phi$$

$$x = r \sin \theta \cos \phi$$

$$= 4 \times \sin 25^\circ \times \sin 120^\circ$$

$$= 4 \times \sin 25^\circ \times \cos 120^\circ$$

$$= 4 \times 0.4226 \times 0.866$$

$$= 4 \times 0.4226 \times (-0.5)$$

$$y = 1.46$$

$$x = -0.845$$

$$z = r \cos \theta$$

$$= 4 \times \cos 25^\circ$$

$$= 4 \times 0.906$$

$$\therefore z = 3.625$$

② Evaluate the divergence of a given vector at $(-1, -1, 2)$.

$$\vec{v} = xy \vec{i} + 2yz \vec{j} + 3zx \vec{k}$$

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sol:

$$\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x} (xy) + \frac{\partial}{\partial y} (2yz) + \frac{\partial}{\partial z} (3zx)$$

$$\nabla \cdot \vec{v} = y + 2z + 3x$$

$$\nabla \cdot \vec{v} \big|_{(-1, -1, 2)} \Rightarrow -1 + (2 \times 2) + 3(-1) = 0$$

∴

It is solenoidal.

③ Determine the divergence of given vector.

$$\vec{v} = x^2 y z \vec{a}_x + x z \vec{a}_z$$

sol:

$$\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$\nabla \cdot \vec{v} \Rightarrow \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (xz)$$

$$\nabla \cdot \vec{v} \Rightarrow 2xyz + x$$

④ Find the divergence of $\vec{Q}$

$$\vec{Q} = \underbrace{\rho \sin \phi}_{Q\rho} \vec{a}_\rho + \underbrace{\rho^2 z}_{Q\phi} \vec{a}_\phi + \underbrace{z \cos \phi}_{Qz} \vec{a}_z$$

sol:

$$\nabla \cdot \vec{Q} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho Q\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} Q\phi + \frac{\partial Qz}{\partial z} \quad \text{⑤}$$

Divergence in cylindrical coordinates $[r, \phi, z]$

$$\nabla \cdot \vec{A} = \frac{1}{r} \frac{\partial (r A_r)}{\partial r} + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

Divergence in spherical coordinates, $[r, \theta, \phi]$

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

Properties:

i) A vector field having its divergence zero is called solenoidal field.

$$\nabla \cdot \vec{A} = 0 \rightarrow \text{Solenoidal.}$$

$$\text{ii) } \nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$$

$$\text{iii) } \nabla \cdot (\nabla A) = \nabla^2 A$$

$$\text{iv) } \nabla \cdot (\vec{A} \vec{B}) = \vec{A} \cdot (\nabla \cdot \vec{B}) + \vec{B} \cdot (\nabla \cdot \vec{A}) \quad \left[\text{used in unit} \right]$$

→ One coil which is formed around a core. Magnetic flux from North to South & South to North. It is in a closed way. No divergence.

Problems

Q. Find the divergence of the given vector.

$$\vec{r} = \frac{5x}{r_x} \vec{i} + \frac{2y}{r_y} \vec{j} - \frac{3z}{r_z} \vec{k}$$

sol:

$$\text{Div}(\vec{r}) = \nabla \cdot \vec{r} \Rightarrow \frac{\partial r_x}{\partial x} + \frac{\partial r_y}{\partial y} + \frac{\partial r_z}{\partial z}$$

$$\Rightarrow \frac{\partial (5x)}{\partial x} + \frac{\partial (2y)}{\partial y} + \frac{\partial (-3z)}{\partial z}$$

$$5 + 2 - 3 = 4 \quad ; \quad \boxed{\nabla \cdot \vec{r} = 4}$$

27/1/25.

[Divergence of vector]

line surface & volume integrals.
Divergence theorems.

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Divergence measures how much the field diverges (or) emerges from the point.

Def:

(scalar) No direction.

→ The divergence of a vector 'A' at a point 'P' is defined as the outward flux per unit volume as the volume shrinks about that point.

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta V \rightarrow 0} \frac{\oint \vec{A} \cdot d\vec{s}}{\Delta V} \quad \text{--- X --- sm.}$$

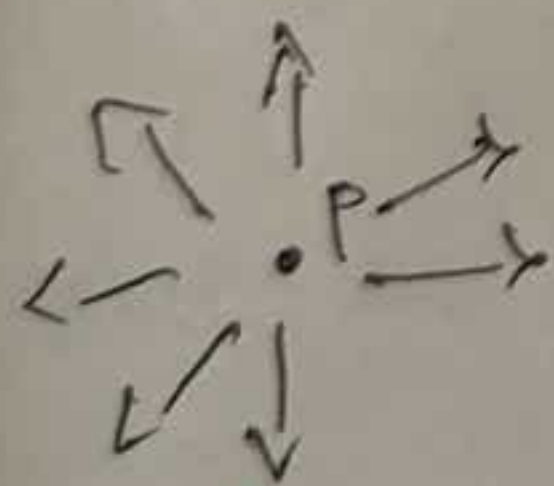
where,

$\oint \vec{A} \cdot d\vec{s} \rightarrow$ net outward flux

$\Delta V \rightarrow$ unit volume.

* The divergence of a vector is a scalar field.

Types of Divergence:



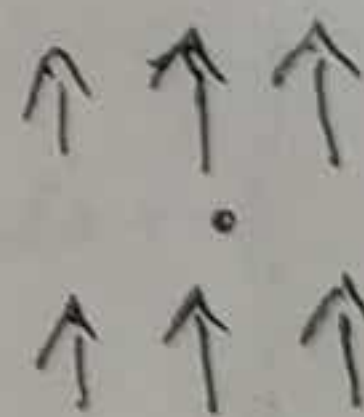
positive Divergence

(source point)



Negative Divergence

(sink point)



Zero Divergence

Divergence in cartesian coordinates, $[x, y, z]$.

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad \text{--- X --- sm.}$$

It is scalar quantity.

total of the conductor is divided into charge, it contains.

$$\nabla \times \nabla \times \vec{H} = \vec{a}_x \frac{\partial}{\partial x} \left\{ \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} \right\} - \vec{a}_x \nabla^2 H_x$$

$$+ \vec{a}_y \frac{\partial}{\partial y} \left\{ \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} \right\} - \vec{a}_y \nabla^2 H_y$$

$$+ \vec{a}_z \frac{\partial}{\partial z} \left\{ \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} \right\} - \vec{a}_z \nabla^2 H_z$$

Divergence of a vector:

$$\nabla \cdot \vec{H} = \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z}$$

$$\nabla \times \nabla \times \vec{H} = \vec{a}_x \frac{\partial}{\partial x} [\nabla \cdot \vec{H}] + \vec{a}_y \frac{\partial}{\partial y} (\nabla \cdot \vec{H})$$

$$+ \vec{a}_z \frac{\partial}{\partial z} (\nabla \cdot \vec{H}) - \nabla^2 [\vec{a}_x H_x +$$

$$\vec{a}_y H_y + \vec{a}_z H_z]$$

$$= \left[\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z} \right] (\nabla \cdot \vec{H}) - \nabla^2 \vec{H}$$

$\nabla \cdot \vec{H}$ valued.

$$\boxed{\nabla \times \nabla \times \vec{H} = \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H}}$$

Hence proved.

of the conductor as it contains

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

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$$\nabla \times \vec{A} \Rightarrow \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \vec{a}_x - \vec{a}_y \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] + \vec{a}_z \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$$

$$\Rightarrow \left[\frac{\partial (xz)}{\partial y} - 0 \right] \vec{a}_x - \vec{a}_y \left[\frac{\partial (xz)}{\partial x} - \frac{\partial (x^2yz)}{\partial z} \right] + \vec{a}_z \left[0 - \frac{\partial (x^2yz)}{\partial y} \right]$$

$$= (0) \vec{a}_x - \vec{a}_y (z - x^2y) + \vec{a}_z [-x^2z]$$

$$\nabla \times \vec{A} \Rightarrow [x^2y - z] \vec{a}_y - x^2z \vec{a}_z$$

2. Determine the curl of the given vector $\vec{F}$ & state the nature of $\vec{F}$.

$$\vec{F} = 30 \vec{a}_x + 2xy \vec{a}_y + 5xz^2 \vec{a}_z \text{ at } (1, 1, -0.2)$$

Sol:

$$F_x = 30 ; F_y = 2xy ; F_z = 5xz^2$$

Curl in cartesian coordinates.

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 30 & 2xy & 5xz^2 \end{vmatrix}$$

$$\nabla \times \vec{F} = \vec{a}_x \left[\frac{\partial (5xz^2)}{\partial y} - \frac{\partial (2xy)}{\partial z} \right] - \vec{a}_y \left[\frac{\partial (5xz^2)}{\partial x} - \frac{\partial (30)}{\partial z} \right] + \vec{a}_z \left[\frac{\partial (2xy)}{\partial x} - \frac{\partial (30)}{\partial y} \right]$$

curl of the conductor is directly
proportional to it contain.

curl in cylindrical coordinates $[r, \phi, z]$.

$$\nabla \times \vec{A} = \frac{1}{r} \begin{vmatrix} \vec{a}_r & r\vec{a}_\phi & \vec{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & rA_\phi & A_z \end{vmatrix}$$

curl in spherical co-ordinates $[r, \theta, \phi]$

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \vec{a}_r & r\vec{a}_\theta & r\sin \theta \vec{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin \theta A_\phi \end{vmatrix}$$

Properties of curl:

The divergence of the curl of a vector
is zero. [Rotational operation]

$$\nabla \cdot (\nabla \times \vec{A}) = 0.$$

The curl of the gradient of scalar field is zero.

$$\nabla \times (\nabla \psi) = 0. \quad (\text{null properties})$$

$\nabla \times \vec{A} = 0 \rightarrow$ curl free field is also called as
Irrotational vector.

$$\boxed{\nabla \times \nabla \times \vec{H} = \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H}} \quad \text{vector identity.}$$

vector identity.

$$\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$$

problem

1. Find the curl of the given vector.

$$\vec{A} = x^2 y z \vec{a}_x + x z \vec{a}_z$$

sol:

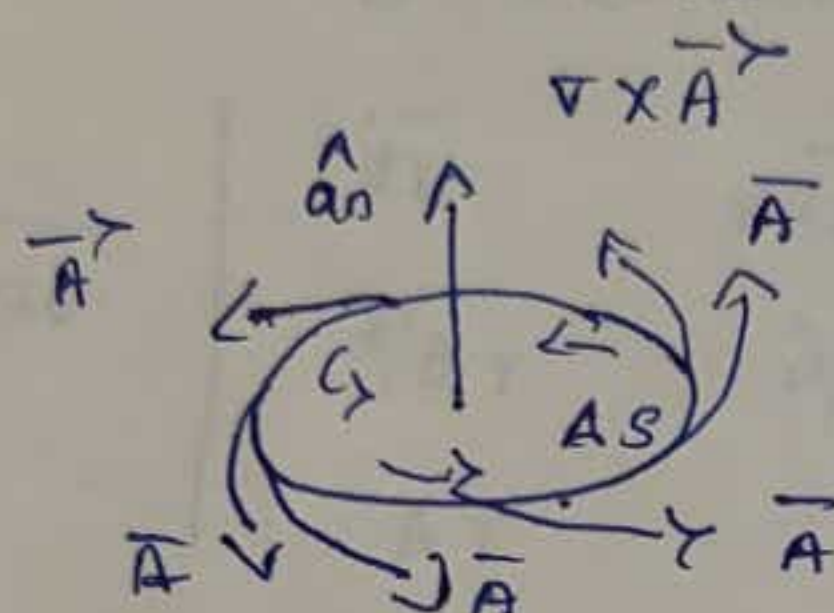
$$A_x = x^2 y z \quad A_y = 0 \quad A_z = x z$$

→ the circulation (or) curl of a vector

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Rotation of a vector field around a closed path is known as curl of a vector → to rotate (or) twist of a vector.

ref:



closed path, 'L'

curl of $\vec{A}$

$$= \nabla \times \vec{A} = \lim_{AS \rightarrow 0} \frac{\oint \vec{A} \cdot d\vec{l}}{AS} \hat{n}$$

$\oint \vec{A} \cdot d\vec{l}$ → circulation of $\vec{A}$ around a closed path.

It can be defined as "the maximum circulation of a vector per unit area as $AS \rightarrow 0$ whose direction is normal to the surface."

Ex:

when a leaf floats in sea water & its rotation is about z-axis (i.e.).

velocity is in z-direction.

* If the curl of a vector is zero, then the vector is irrotational vector. $[\nabla \times \vec{A} = 0]$.

curl in cartesian co-ordinate system $[x, y, z]$.

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\vec{A} = A_x \vec{a}_x + A_y \vec{a}_y + A_z \vec{a}_z$$

$$\nabla \times \vec{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \vec{a}_x - \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] \vec{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \vec{a}_z$$

central of the conductor is directly
in line with it contain

$$\nabla \cdot \vec{Q} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \times \rho \sin \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (\rho^2 z) + \frac{\partial}{\partial z} [z \cos \phi]$$

ρ^2
 $\downarrow$
 dist

$$\Rightarrow \frac{1}{\rho} [2 \rho \sin \phi] + \frac{1}{\rho} (0) + \cos \phi$$

$$\therefore \nabla \cdot \vec{Q} = 2 \sin \phi + \cos \phi.$$

(5) Find the divergence of a vector.

$$\vec{F} = \underbrace{\frac{1}{r^2} \cos \theta}_{F_r} \vec{a}_r + \underbrace{r \sin \theta \cos \phi}_{F_\theta} \vec{a}_\theta + \underbrace{\cos \phi}_{F_\phi} \vec{a}_\phi$$

sol:

$$\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (F_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (F_\phi)$$

$$\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \times \frac{1}{r^2} \cos \theta \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin \theta \cos \phi \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (\cos \theta)$$

$\frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (\cos \theta)$

$$= 0 + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin^2 \theta \cos \phi) + 0 = \frac{r \cos \phi}{r \sin \theta} [2 \sin \theta \cos \theta]$$

$$= \frac{r \cos \phi}{r \sin \theta} 2 \sin \theta \cos \theta$$

$$\boxed{\nabla \cdot \vec{F} = 2 \sin \theta \cos \theta \cos \phi}$$

$$\left| \frac{\partial}{\partial \theta} (\sin^2 \theta) = 2 \sin \theta \cos \theta \right|$$

$$= \int_0^1 3 dx \Rightarrow (3x)_0^1 = 3 - 0 = 3.$$

$$\iiint_V \nabla \cdot \vec{A} \, dV = 3 \quad \text{--- (1)}$$

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R.H.S

surface integral of $\vec{A}$.

$$\oiint \vec{A} \cdot d\vec{S} = \iint A \cdot \vec{i} \, dy \, dz + \iint A \cdot (-\vec{i}) \, dy \, dz + \iint A \cdot \vec{j} \, dx \, dz + \iint A \cdot (-\vec{j}) \, dx \, dz + \iint A \cdot \vec{k} \, dx \, dy + \iint A \cdot (-\vec{k}) \, dx \, dy \quad \text{--- (2)}$$

$$\iint A \cdot \vec{i} \, dy \, dz = \int_0^1 \int_0^1 (4xz \vec{i} - 2y^2 \vec{j} + z^2 \vec{k}) \cdot \vec{i} \, dy \, dz$$

$$[\because \vec{i} \cdot \vec{i} = 1; \vec{i} \cdot \vec{j} = 0]$$

↓ dot product.

$$= \int_0^1 \int_0^1 4xz \vec{i} \cdot \vec{i} - 2y^2 \vec{j} \cdot \vec{i} + z^2 \vec{k} \cdot \vec{i} \, dy \, dz.$$

$$= \int_0^1 \int_0^1 [4xz] \, dy \, dz = \int_0^1 (4xz)_0^1 \, dz$$

$$= \int_0^1 4xz \, dz = [4xz]_0^1 = 4x$$

$$\Rightarrow \int_0^1 \int_0^1 A \cdot \vec{i} \, dy \, dz = 4x$$

since. $x=1$ in $\vec{i}$;

$$\iint A \cdot \vec{i} \, dy \, dz = 4.$$

$$\iint A \cdot (-\vec{i}) \, dy \, dz = 0 \quad [\because x=0 \text{ in } -\vec{i}]$$

$$\Rightarrow \text{for } 4x \vec{i} \cdot (-\vec{i}) \Rightarrow -4x$$

lower line 0
max line 1

$$[-\vec{i}] \rightarrow$$

$$\ast \iint A \cdot \vec{j} \, dx \, dy = \int_0^1 \int_0^1 [4xz \vec{i} - 2y^2 \vec{j} + z^2 \vec{k}] \cdot \vec{j} \, dx \, dz$$

problems in divergence Theorem.

1. state and verify divergence theorem for the vector.
 $\vec{A} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ taken over the cube bounded by
 $x=0, x=1, y=0, y=1$ & $z=0, z=1$.

Sol:

statement:

$$\iiint_V \nabla \cdot \vec{A} \, dv = \oint_S \vec{A} \cdot n \, dS$$

The volume integral of the divergence of a vector field over a volume is equal to the surface integral of the normal component of this vector over the surface bounding the volume.

*. verify the Divergence theorem:

Given: $\vec{A} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$.

L.H.S

Taking the Divergence.

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x} (4x) + \frac{\partial}{\partial y} (-2y^2) + \frac{\partial}{\partial z} (z^2)$$

$$\nabla \cdot \vec{A} = 4 - 4y + 2z$$

Taking volume integral over V .

$$\iiint_V \nabla \cdot \vec{A} \, dv = \int_0^1 \int_0^1 \int_0^1 (4 - 4y + 2z) \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^1 \left[\int_0^1 (4 - 4y + 2z) \, dz \right] dx \, dy$$

$$= \int_0^1 \int_0^1 \left[4z - 4yz + \frac{2z^2}{2} \right]_0^1 dx \, dy$$

$$\iiint_V \nabla \cdot \vec{A} \, dv = \int_0^1 \int_0^1 (4 - 4y + 1) \, dx \, dy$$

$$= \int_0^1 \left[\int_0^1 (5 - 4y) \, dy \right] dx$$

$$= \int_0^1 \left[5y - \frac{4y^2}{2} \right]_0^1 dx = \int_0^1 (5 - 2) \, dx$$

of the conductor is
range, it contain.

Find the constants a, b & c such that the given vector is irrotational.

$$\vec{A} = (x+2y+az)\vec{a}_x + (bx-3y-z)\vec{a}_y + (4x+cy+2z)\vec{a}_z$$

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Sol:

curl in cartesian coordinates.

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \vec{a}_x \frac{\partial F_z}{\partial y} - \vec{a}_y \frac{\partial F_z}{\partial x} + \vec{a}_z \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

Given.

$$A_x = x+2y+az ; A_y = bx-3y-z ; A_z = 4x+cy+2z$$

Irrotational vector: $\nabla \times \vec{A} = 0$.

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = 0$$

$$\vec{a}_x \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] - \vec{a}_y \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] + \vec{a}_z \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$$

$$\left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] = \left[\frac{\partial A_z}{\partial x} + \frac{\partial A_x}{\partial z} \right] = \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] = 0.$$

$$i) \frac{\partial (4x+cy+2z)}{\partial y} = \frac{\partial (bx-3y-z)}{\partial z} = 0$$

$$c+1=0$$

$$\boxed{c=-1}$$

$$ii) \frac{\partial (x+2y+az)}{\partial z} = \frac{\partial (4x+cy+2z)}{\partial x} = 0$$

$$a-4=0 \quad a=4$$

$$iii) \frac{\partial (bx-3y-z)}{\partial x} = \frac{\partial (x+2y+az)}{\partial y} = 0$$

$$b-2=0 \Rightarrow b=2$$

$$\Rightarrow \bar{a}_x [0 - 0] - \bar{a}_y [5z^2 - 0] + \bar{a}_z [2y - 0]$$

$$\nabla \times \vec{F} = -5z^2 \bar{a}_y + 2y \bar{a}_z$$

curl at point $(1, 1, -0.2)$

$$\nabla \times \vec{F} = -5(-0.2)^2 \bar{a}_y + 2(1) \bar{a}_z$$

$$\nabla \times \vec{F} = 0.2 \bar{a}_y + 2 \bar{a}_z \quad (0.04)$$

$\therefore$ As the curl is not zero, the given vector is rotational in nature.

3. Determine the curl of given vector:

$$\vec{A} = \rho \sin \phi \bar{a}_\rho + \rho^2 z \bar{a}_\phi + z \cos \phi \bar{a}_z$$

Sol:

Given: $A_\rho = \rho \sin \phi$; $A_\phi = \rho^2 z$; $A_z = z \cos \phi$.

curl in cylindrical coordinates.

$$\nabla \times \vec{A} = \frac{1}{\rho} \begin{vmatrix} \bar{a}_\rho & \rho \bar{a}_\phi & \bar{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$= \frac{1}{\rho} \left\{ \bar{a}_\rho \left[\frac{\partial A_z}{\partial \phi} - \frac{\partial (\rho A_\phi)}{\partial z} \right] - \rho \bar{a}_\phi \left[\frac{\partial A_z}{\partial \rho} - \frac{\partial A_\rho}{\partial z} \right] + \bar{a}_z \left[\frac{\partial (\rho A_\phi)}{\partial \rho} - \frac{\partial A_\phi}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \right\}$$

$$= \frac{1}{\rho} \left\{ \bar{a}_\rho \left[\frac{\partial (z \cos \phi)}{\partial \phi} - \frac{\partial (\rho \rho^2 z)}{\partial z} \right] - \rho \bar{a}_\phi \left[\frac{\partial (z \cos \phi)}{\partial \rho} - \frac{\partial (\rho \sin \phi)}{\partial z} \right] + \bar{a}_z \left[\frac{\partial (\rho \rho^2 z)}{\partial \rho} - \frac{\partial (\rho \sin \phi)}{\partial \phi} \right] \right\}$$

$$= \frac{1}{\rho} \left\{ \bar{a}_\rho [z(-\sin \phi) - \rho^3] - \rho \bar{a}_\phi [0 - 0] + \bar{a}_z [3\rho^2 z - \rho \cos \phi] \right\}$$

$$\Rightarrow \left[-\frac{z \sin \phi}{\rho} - \rho^2 \right] \bar{a}_\rho + [3\rho z - \cos \phi] \bar{a}_z$$

(Stoke's theorem)

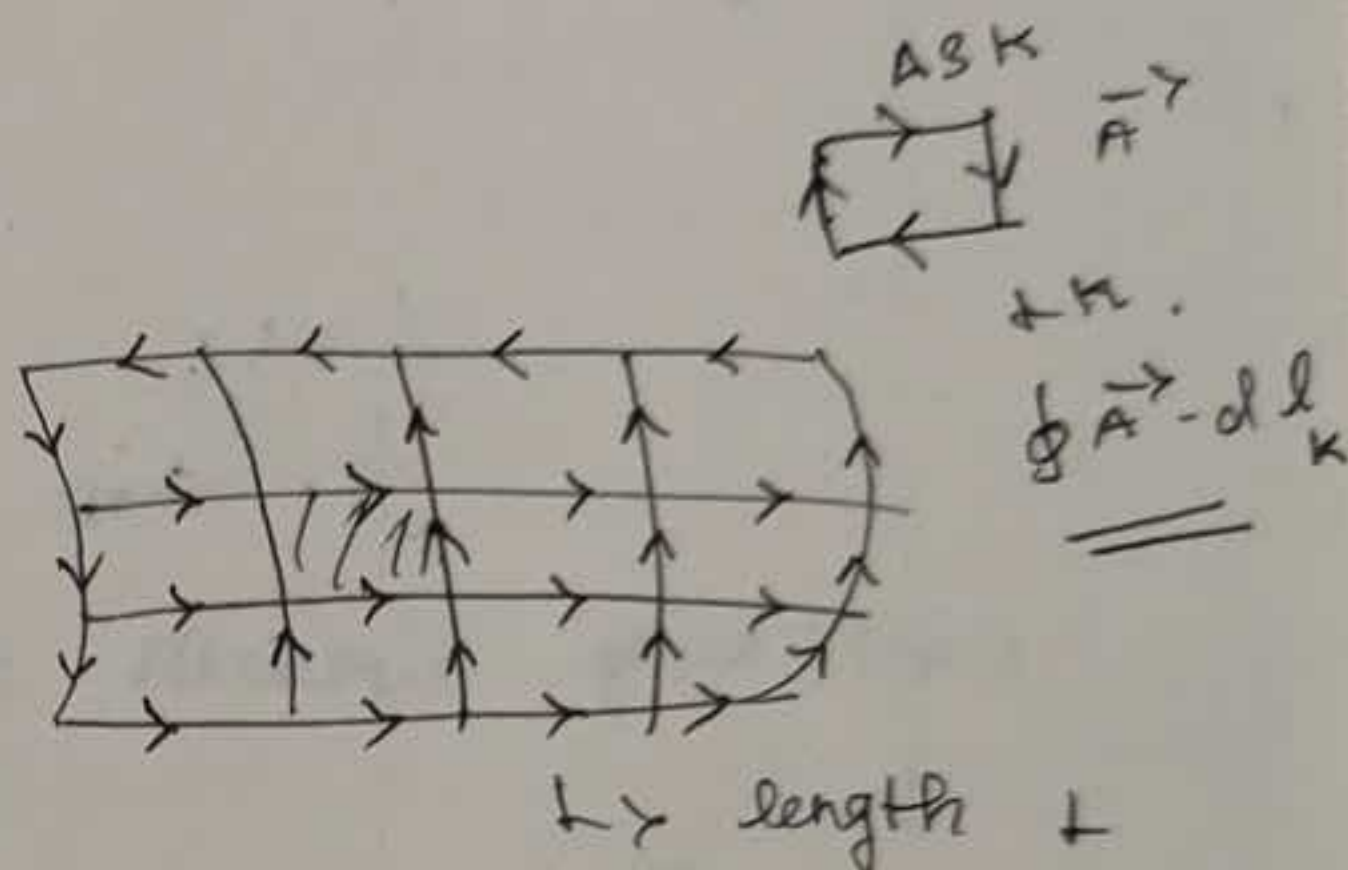
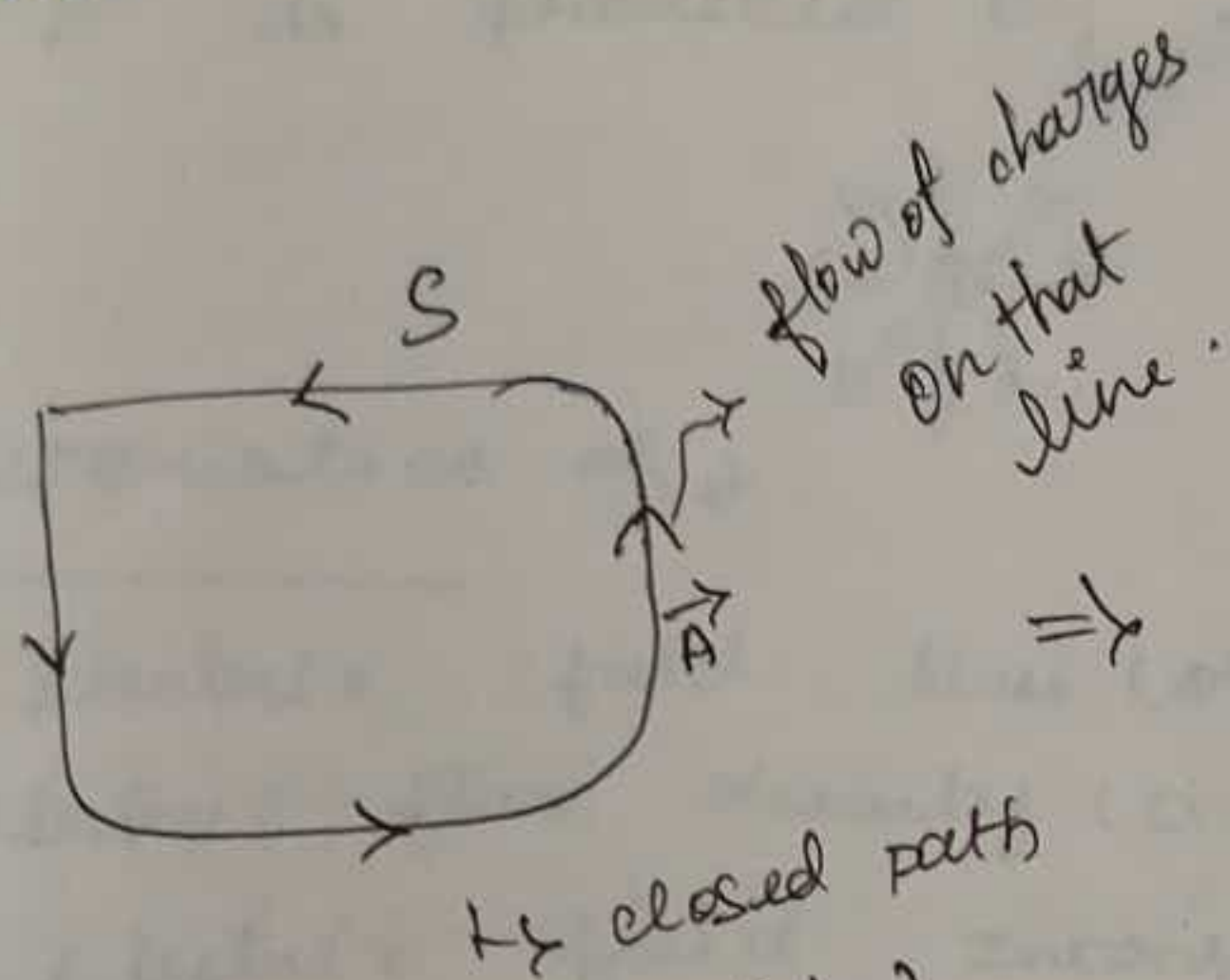
It relates line integral & surface integral.

Defn:

$$\oint_L \vec{H} \cdot d\vec{l} = \iint_S (\nabla \times \vec{H}) \cdot d\vec{s}$$

The line integral of a vector around a closed path is equal to the surface integral of the normal component of its curl over any closed surface.

Proof:



[circulation of vector] (L) $\rightarrow$ surface

$\rightarrow$ surface 'S' is subdivided into 'n' no. of cells.

$\rightarrow$ let k^{th} cell has surface area ΔS_k bounded by L_k .

$$\oint_{L_k} \vec{A} \cdot d\vec{l} \approx \text{circulation of } \vec{A} \text{ around } L_k.$$

$\therefore$ For total closed path,

$$\oint_L \vec{A} \cdot d\vec{l} = \sum_k \oint_{L_k} \vec{A} \cdot d\vec{l}$$

$\times \Delta$ this smaller surface area. (ΔS_k).

$$\oint_L \vec{A} \cdot d\vec{l} = \sum_k \left(\frac{\oint_{L_k} \vec{A} \cdot d\vec{l}}{\Delta S_k} \right) \Delta S_k.$$

$$= \sum_k (\nabla \times \vec{A}) \cdot \Delta S_k.$$

$$\oint_L \vec{A} \cdot d\vec{l} = \iint_S \nabla \times \vec{A} \cdot d\vec{s} \quad \text{hence proved the Stoke's theorem.}$$

$$\iint \vec{D} \cdot \vec{a}_x dy dz = 12x = 12$$

$$\iint \vec{D} \cdot (-\vec{a}_x) dy dz = -12x = 0 \quad \left. \begin{array}{l} x=1 \\ x=0 \end{array} \right\} \text{--- (3)}$$

$$* \iint \vec{D} \cdot \vec{a}_y dx dz :$$

$$\Rightarrow \int_0^1 \int_0^3 [2xy \vec{a}_x + x^2 \vec{a}_y] \cdot \vec{a}_y dx dz$$

$$= \int_0^1 \int_0^3 [x^2] dx dz$$

$$= \int_0^1 [x^2 z]_0^3 dz = \int_0^1 9z dz$$

$$= \int_0^1 3x^2 dx = \left[\frac{3x^3}{3} \right]_0^1 = 1.$$

$$\iint \vec{D} \cdot \vec{a}_y dx dz = 1 \quad \left. \begin{array}{l} \text{--- (4)} \\ \text{--- (5)} \end{array} \right\} \begin{array}{l} \text{if there mean.} \\ \text{we can use.} \\ \text{NO } y \text{ term.} \end{array}$$

$$\iint \vec{D} \cdot (-\vec{a}_y) dx dz = -1$$

$$[\vec{a}_x \cdot \vec{a}_z = \vec{a}_y \cdot \vec{a}_z = 0]$$

$\Rightarrow$

$$* \iint \vec{D} \cdot \vec{a}_z dx dy :$$

$$\Rightarrow \int_0^1 \int_0^2 [2xy \vec{a}_x + x^2 \vec{a}_y] \cdot \vec{a}_z dx dy = 0.$$

$$\iint \vec{D} \cdot \vec{a}_z dx dy = \iint \vec{D} \cdot (-\vec{a}_z) dx dy = 0 \quad \text{--- (5)}$$

substitute (3), (4) & (5) in (2)

$$\oiint \vec{D} \cdot d\vec{s} = 12 + 0 + 1 - 1 + 0 = 12 \quad \text{--- (6)}$$

(From)

$$1 \leq 2 \quad \iiint_V r \cdot r dr = \oiint \vec{D} \cdot d\vec{s} = 12 //$$

Given
 $\vec{D} = 2xy\vec{a}_x + x^2\vec{a}_y$

$x=0, x=1; y=0, y=2; z=0, z=3.$

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Divergence Theorem:

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$$\iiint_V \nabla \cdot \vec{D} \, dv = \oiint_S \vec{D} \cdot \hat{n} \, ds.$$

L.H.S

Taking the divergence of the given vector.

$$\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\nabla \cdot \vec{D} = \frac{\partial}{\partial x} [2xy] + \frac{\partial}{\partial y} [x^2]$$

$$\nabla \cdot \vec{D} = 2y.$$

Taking the volume integral on side.

$$\iiint_V \nabla \cdot \vec{D} \, dv = \iiint_0^3 \int_0^2 \int_0^1 [2y] \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^2 [2yz]_0^1 \, dy \, dz = \int_0^1 \int_0^2 2yz \, dy \, dz = \int_0^1 \left[\frac{2y^2}{2} \right]_0^2 \, dz$$

$$= \int_0^1 [2y^2]_0^2 \, dz = \int_0^1 4z \, dz = \left[\frac{4z^2}{2} \right]_0^1 = 2.$$

R.H.S

To find $\oiint_S \vec{D} \cdot d\vec{s}$

$$\oiint_S \vec{D} \cdot d\vec{s} = \iint_{x=1} \vec{D} \cdot \vec{a}_x \, dy \, dz + \iint_{x=0} \vec{D} \cdot (-\vec{a}_x) \, dy \, dz$$

$$+ \iint_{y=2} \vec{D} \cdot \vec{a}_y \, dx \, dz + \iint_{y=0} \vec{D} \cdot (-\vec{a}_y) \, dx \, dz$$

$$+ \iint_{z=3} \vec{D} \cdot \vec{a}_z \, dx \, dy + \iint_{z=0} \vec{D} \cdot (-\vec{a}_z) \, dx \, dy$$

$$\iint_{x=1} \vec{D} \cdot \vec{a}_x \, dy \, dz \Rightarrow \iint (2xy\vec{a}_x + x^2\vec{a}_y) \cdot \vec{a}_x \, dy \, dz$$

$$= \int_0^2 \int_0^3 (2xy) \, dy \, dz$$

$$\left[\because \vec{a}_x \cdot \vec{a}_x = 1 \right. \\ \left. \vec{a}_y \cdot \vec{a}_x = 0 \right]$$

$$[y=0 \text{ to } 2], [z=0 \text{ to } 3]$$

$$\iint_{x=1} \vec{D} \cdot \vec{a}_x \, dy \, dz = \int_0^2 (2xy) \, dy = \int_0^2 (2xy) \, dy$$

$$= \left[\frac{2xy^2}{2} \right]_0^2 = 2xy^2 \Big|_0^2 = 12x = 12$$

$$x=1$$

$$x=0$$

$$\Rightarrow \int_0^1 \int_0^1 (-2y^2) dx dz \Rightarrow \int_0^1 (-2y^2 z)_0^1 dx$$

$$= -2 \int_0^1 y^2 dx \Rightarrow -2 (y^2 x)_0^1 \Rightarrow -2y^2$$

$$\iint \vec{A} \cdot \vec{j} dx dy = -2y^2.$$

since $y=1$ in $\vec{j}$, $\iint \vec{A} \cdot \vec{j} dx dz = -2$

$\iint \vec{A} \cdot (-\vec{j}) dx dz = 0$ [$y=0$ in $-\vec{j}$]

$$* \iint \vec{A} \cdot \vec{k} dx dy = \int_0^1 \int_0^1 (4xi + 2y^2j + z^2k) \cdot \vec{k} dx dy.$$

$$= \int_0^1 \int_0^1 z^2 dx dy = \int_0^1 (z^2 y)_0^1 dy = \int_0^1 z^2 dz$$

$$= z^2$$

since: $z=1$ in $\vec{k}$, $\iint \vec{A} \cdot \vec{k} dx dy = 1$

$\iint \vec{A} \cdot (-\vec{k}) dx dy = 0$

substitute.

3, 4, & 5 in (2).

$$\oiint \vec{A} \cdot d\vec{S} \Rightarrow 4+0+(1-2)+0+(1+0)$$

$$= 5-2=3.$$

$$\therefore \boxed{\oiint \vec{A} \cdot d\vec{S} = 3} \quad (6)$$

compare (1) & (6).

$$\iiint_V \nabla \cdot \vec{A} \cdot dV = \oiint \vec{A} \cdot d\vec{S} = 3.$$

(2). check validity of the divergence theorem considering the field $\vec{D} = 2xy \vec{a}_x + x^2 \vec{a}_y$ c/m² & the rectangular parallelepiped formed by the planes. $x=0, x=1, y=0, y=2, z=0$ & $z=3$.

I-A.

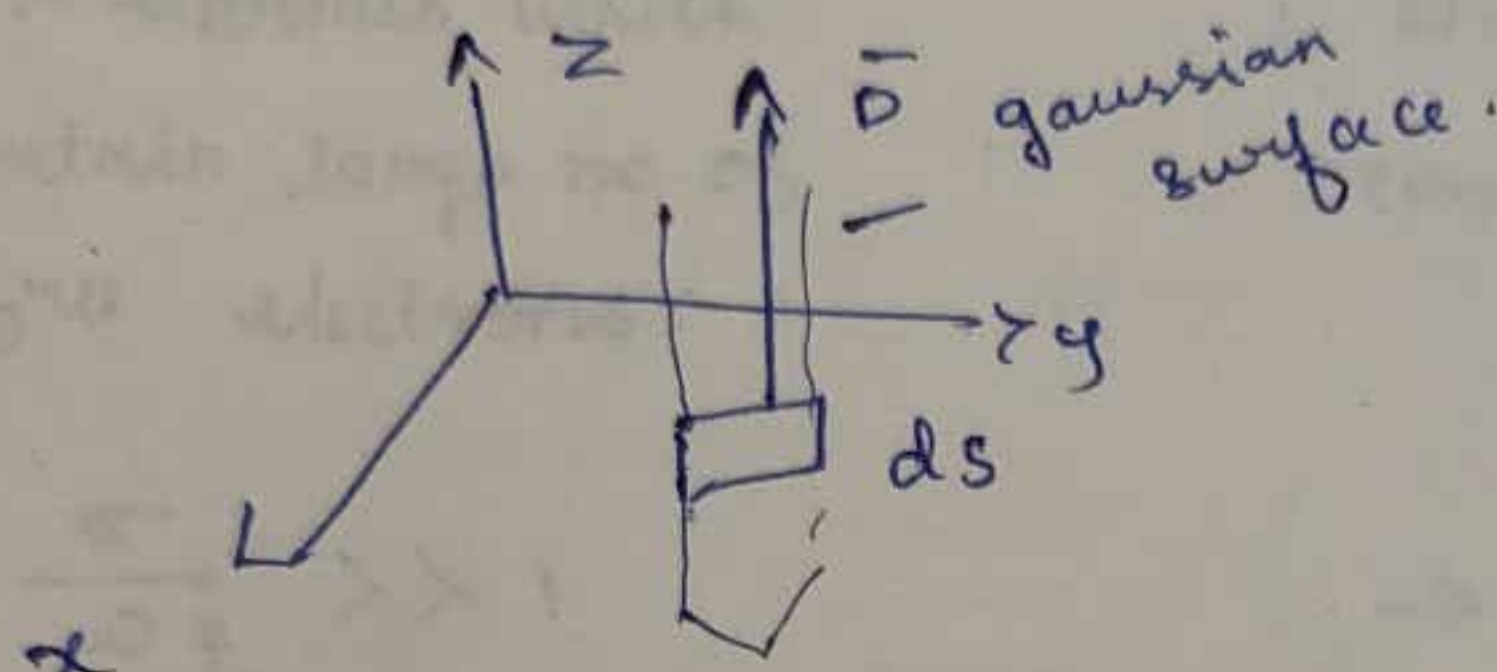
Infinite sheet of charge.

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$$Q = \int \vec{D} \cdot d\vec{s} \quad (1)$$



$$Q = \int_{\text{top}} \vec{D} \cdot d\vec{s} + \int_{\text{bottom}} \vec{D} \cdot d\vec{s} + \int_{\text{lateral}} \vec{D} \cdot d\vec{s}$$

lateral $\downarrow 0$

$$Q = \int_{\text{top}} D_z \vec{a}_z \cdot d\vec{s} + \int_{\text{bottom}} D_z (-\vec{a}_z) \cdot d\vec{s}$$

$$Q = \int D_z dxdy + \int D_z dxdy$$

$$Q = D_z \int dxdy + D_z \int dxdy$$

$$Q = D_z A + D_z A$$

$$Q = 2 D_z A$$

$$D_z = \frac{Q}{2A}$$

$\boxed{PS} \rightarrow A$

$$Q = PSA$$

$$D_z = \frac{PSA}{2A}$$

$$\vec{D} = D_z \vec{a}_z$$

$$\boxed{\vec{D} = \frac{PS}{2} \vec{a}_z}$$

C/m²

$$\vec{D} = \epsilon_0 \vec{E}$$

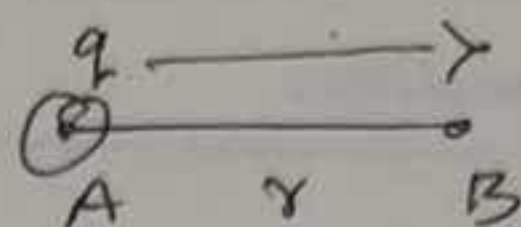
$$\vec{E} = \frac{PS \vec{a}_z}{2\epsilon_0}$$

V/m

Electric potential: (V)

It is the amount of work required to move a unit of charge from one point to another in an electric field.

$$V \propto \frac{q}{r}$$



2m

$$V = \frac{q}{4\pi\epsilon_0 r}$$

$$; \text{ volts } \leftarrow V \rightarrow$$

charge ↑
potential ↑

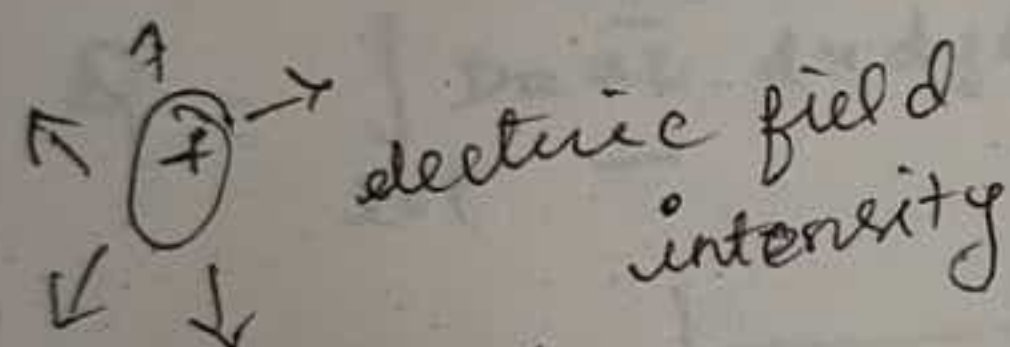
* Relation between (V) & (E) : -

$$E = -\nabla V$$

$$V = -\int E \cdot dl$$

Electric field is the gradient of voltage. voltage is the integral of electric field.

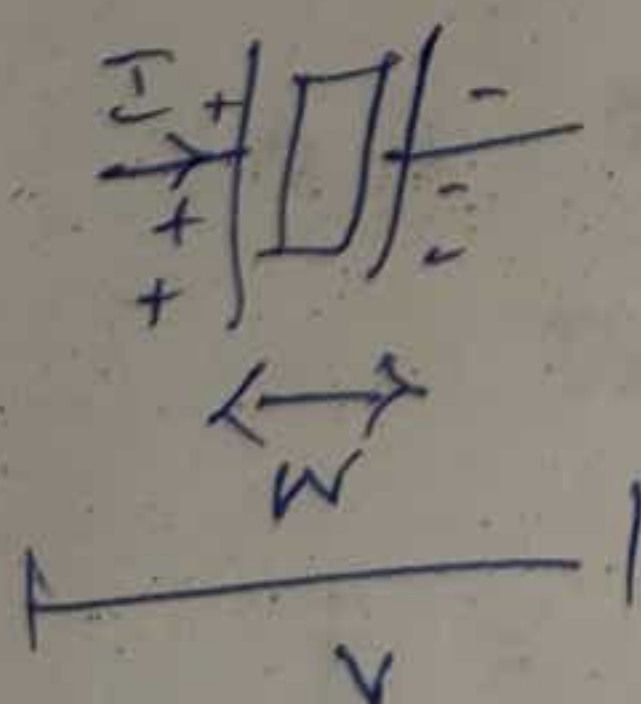
'-' sign indicates that the direction of $\vec{E}$ is opposite to 'V'.



Electric potential.

Electrostatic Energy (W):

It is defined as the work done to build up the charge in capacitor. capacitor stores electrostatic energy.



$$W = \frac{1}{2} CV^2 ; \text{ joules.}$$

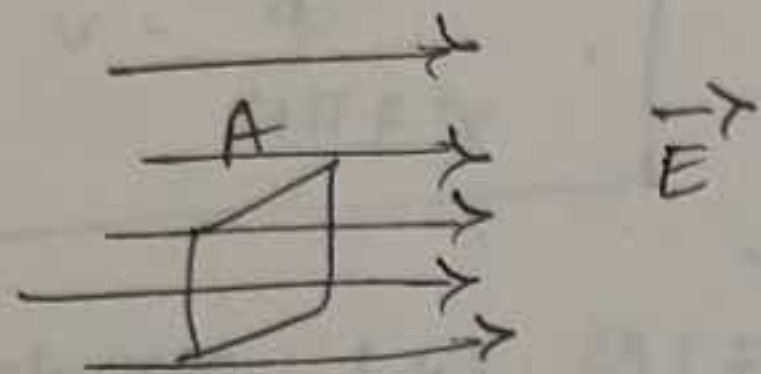
$$W = \frac{1}{2} QV \text{ where } C = \frac{Q}{V}$$

$$C = \frac{\text{charge} \times \text{voltage}}{\text{capacitance}}$$

Electric Flux density: (D) :-

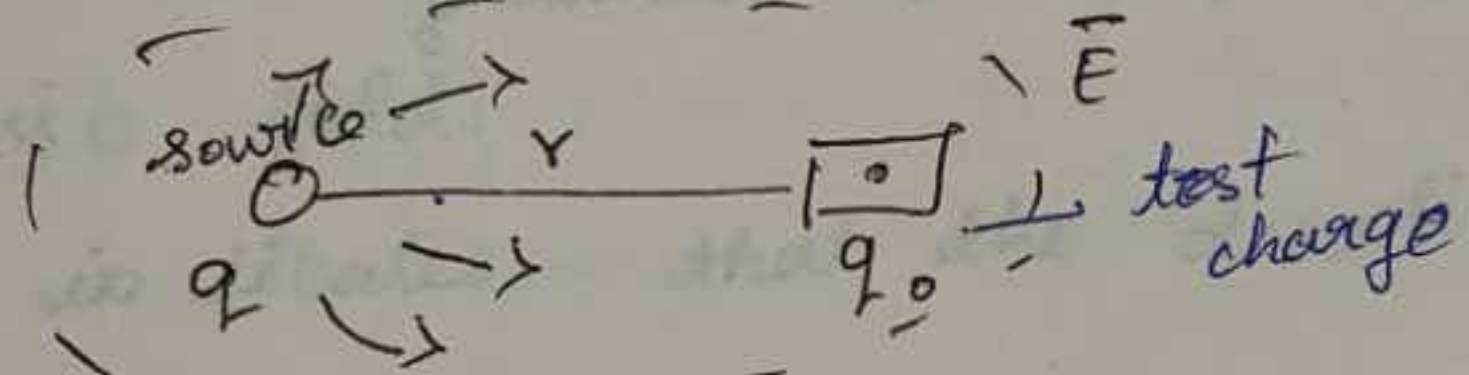
It is defined as the amount of flux passing through unit surface area perpendicular to the direction of the flux.

$$D = \Psi / A ; \text{ coulomb/m}^2.$$



Electric field Intensity (E) :-

It is the measure of the strength of an electric field at any point.



depends on distance & force.

It is defined as the electric force per unit charge experienced by the charge which is placed in the electric field.

$$E = \frac{F_{\text{force}}}{q_0 \text{ unit charge}} \text{ Newtons per coulomb (or)}.$$

$$E = \frac{q}{4\pi\epsilon_0 r^2} ; \text{ volt/m.}$$

Inversely proportional to the distance. $\uparrow$ Intensity $\downarrow$ Electric field.

Relation b/w Ψ , D & E :-

Flux density:

$$D = \frac{\Psi}{A}$$

$$\Psi = D \cdot A.$$

$$\Psi = \int D \cdot dS \Rightarrow D \cdot A.$$



$$\therefore \boxed{D = \epsilon E}$$

$$\epsilon \Rightarrow \epsilon_0 \epsilon_r$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$\boxed{D \propto E}$$

of the conductor is directly proportional to the charge it contains.

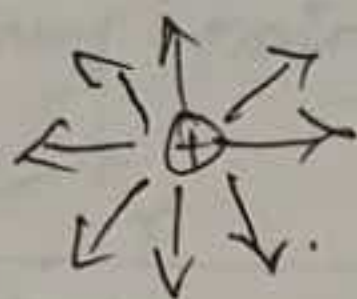
for sphere coordinates system
 $\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial V}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial V}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$

uses:

- It is mainly used to convert the surface integral into line integral and vice versa.
- To find line integral of some particular curve.
- To determine the curl of a bounded surface.

Electrostatics → static Electric Field.

- It is the study of charge at rest condition.
- $\vec{E}$ is produced by the stationary charge.

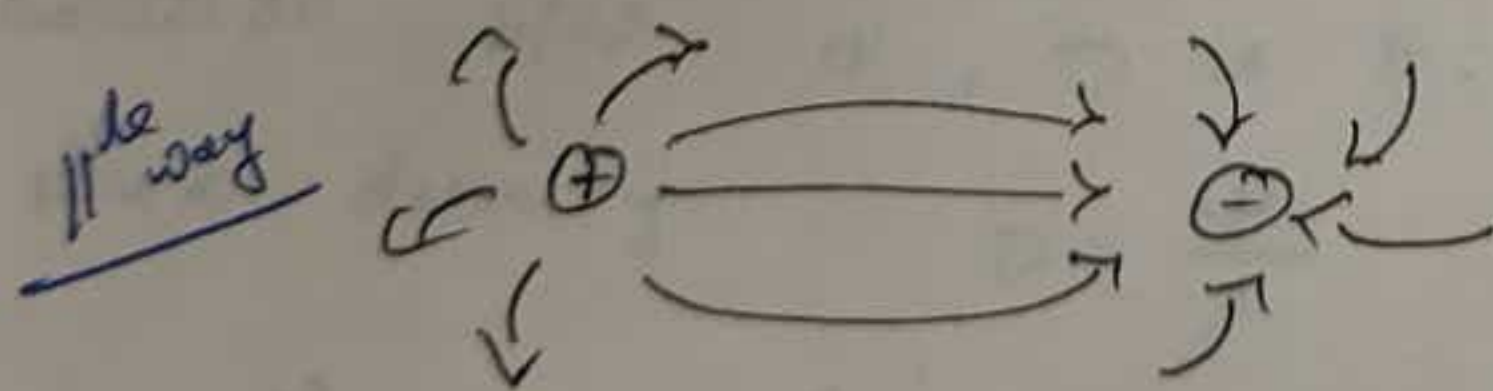


characteristics of $\vec{E}$.

- Electric field lines (or) Electric flux (ψ)
- Electric flux density (D)
- Electric field intensity (E)
- Electric potential (V)

Electric field lines:

- Electric field lines are imaginary lines which give virtual idea of electric field.
- It can never be a closed loop.

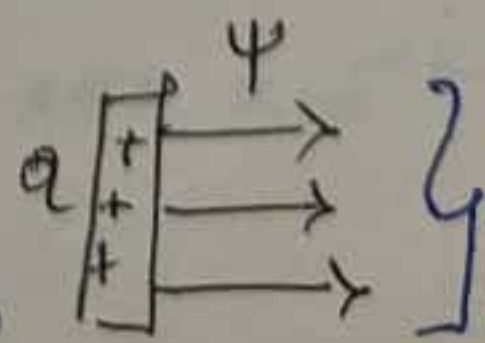


- Electric field lines never cross each other.

- It can never be a closed loop.

Electric flux (ψ):-

- It is the total number of electric lines of force emanating from a charge.



- If Q is the charge of body, then the electric flux produced from that is given as,

$$\psi = Q$$

$\nabla^2 v = 0$: The solutions of Laplace's equation are the harmonic function.

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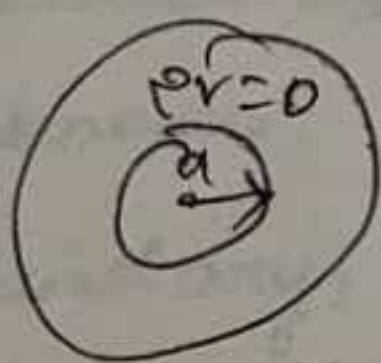
→ Harmonic functions are important in partial differential equation theory, physics & engineering.

Applications:

→ it is used in boundary value problem.

Ex:

TO obtain potential & capacitance between 2 surfaces.



Coaxial cable.

potential b/w conductors of coaxial cable is

$$\nabla^2 v = 0.$$

Poisson:

$$\nabla^2 v = \frac{\rho v}{\epsilon}$$

Laplace:

$$\nabla^2 v = 0$$

-X-

sub (1) in (3)

$$\nabla(-\nabla v) = \frac{\rho v}{\epsilon}$$

$$-\nabla^2 v = \frac{\rho v}{\epsilon} \Rightarrow \boxed{\nabla^2 v = -\frac{\rho v}{\epsilon}} \quad \text{---X}$$

$\Rightarrow$ Poisson's equation in coordinate systems:

$$\nabla^2 v = -\frac{\rho v}{\epsilon}$$

for cartesian co-ordinate system (x, y, z)

$$\nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = -\frac{\rho v}{\epsilon}$$

for cylindrical co-ordinate system (ρ, ϕ, z)

$$\nabla^2 v = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 v}{\partial \phi^2} \right) + \frac{\partial^2 v}{\partial z^2} = -\frac{\rho v}{\epsilon}$$

for spherical (r, θ, ϕ) .

$$\nabla^2 v = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 v}{\partial \phi^2} = -\frac{\rho v}{\epsilon}$$

Laplace's equation.

If an enclosed surface does not have any volume charge (ρv), then the Poisson's equation becomes.

$$\rho v = Q/v$$

$$\underline{\rho v = 0}$$

$$\boxed{\nabla^2 v = 0}$$

This is called as Laplace

equation.

$\nabla^2 \rightarrow$ Laplacian operator.

→ Poisson's equation & Laplace's equation:

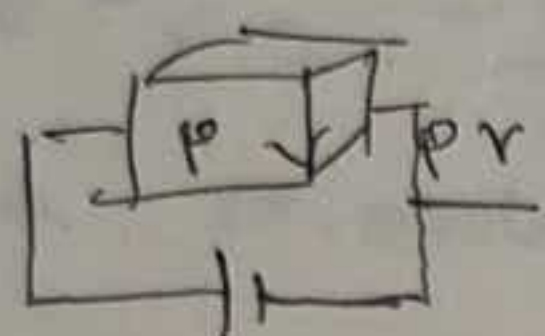
Poisson's Equation:

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Poisson's equation relates the electric potential (V) to charge density.



volume charge density

$$\rho_v = \frac{Q - \text{charge}}{V - \text{volume}}$$

$$\nabla^2 V = - \frac{\rho_v}{\epsilon}$$

where.

$\nabla^2 \Rightarrow$ Laplacian operator.

$V \rightarrow$ electric potential.

$\rho_v \rightarrow$ volume charge density.

$\epsilon \rightarrow$ dielectric constant.

→ The potential of a conductor is directly proportional to the charge it contains. $[V \propto \rho_v]$

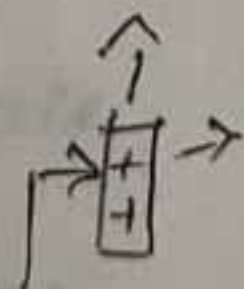
Proof:-

The relation between V & E is given as.

$\Rightarrow$ gradient of electric potential in opposite direction.

$$E = -\nabla V$$

— (1)



$\downarrow$ Electric field intensity.

→ Gauss's law in point form.

$$\nabla \cdot D = \rho_v$$

$$\nabla \cdot (E\epsilon) = \rho_v$$

$$\nabla \cdot E = \frac{\rho_v}{\epsilon}$$

— (3)

$$\nabla \cdot D = \rho_v$$

— (2)

$$\oint \phi = \frac{Q}{\epsilon_0}$$

$$\oint D \cdot ds = Q$$

$$\nabla \cdot D = \rho_v$$

$$D \propto E$$

$$\Rightarrow E$$

unit 2

conductors in static electric field:

conductors

→ materials which contain large no. of free electrons.

$$\rightarrow \frac{\sigma}{\omega \epsilon} \gg 1$$

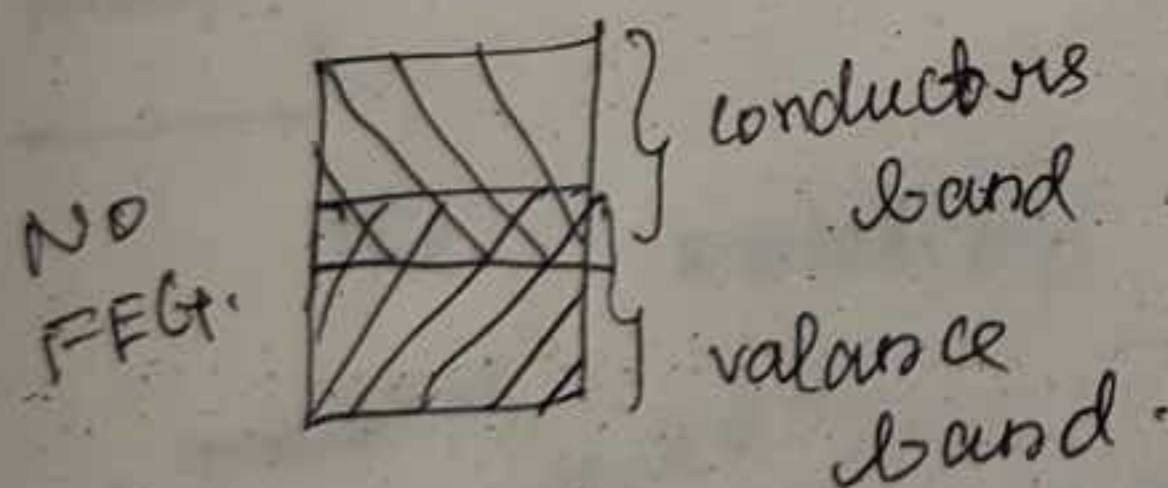
where.

$\sigma \rightarrow$ conductivity.

$\epsilon \rightarrow$ dielectric constant.

→ Excellent conductivity
Ex: metals, iron, copper, silver etc.

→ No forbidden energy gap.



→ low electrical resistance.

* Negative temperature coefficient.

i.e) conductivity decreases with temperature rise.

dielectric (insulator)

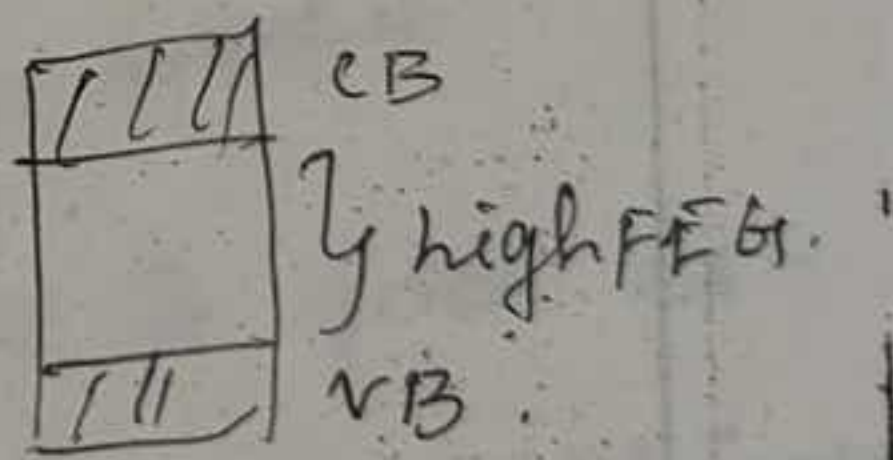
1. materials which do not contain free electrons.

$$\rightarrow \frac{\sigma}{\omega \epsilon} \ll 1$$

conductivity is always less than permittivity.

→ poor conductivity.
Ex: Non metals.
wood, rubber, plastic, etc.

→ very high FEG.



→ very high electrical resistance.

* positive temperature coefficient conductivity increases with temperature rise.

The tangential component of 'E'

$$E_{t1} = E_1 \sin \theta_1$$

$$E_{t2} = E_2 \sin \theta_2 \quad \text{கருதல் பத்தகம்}$$

Ed. B-132

பரிகிவரண...

Apply Boundary conditions with $P_S = 0$

$$E_{t1} = E_{t2} \quad \& \quad D_{n1} = D_{n2}$$

$$E_1 \sin \theta_1 = E_2 \sin \theta_2 \quad \text{--- (1)}$$

$$D_1 \cos \theta_1 = D_2 \cos \theta_2 \quad \text{--- (2)}$$

$$\rightarrow \textcircled{1} \div \textcircled{2}$$

$$\frac{E_1 \sin \theta_1}{D_1 \cos \theta_1} \Rightarrow \frac{E_2 \sin \theta_2}{D_2 \cos \theta_2}$$

$$\frac{E_1}{\epsilon_1 E_1} \tan \theta_1 \Rightarrow \frac{E_2}{\epsilon_2 E_2} \tan \theta_2$$

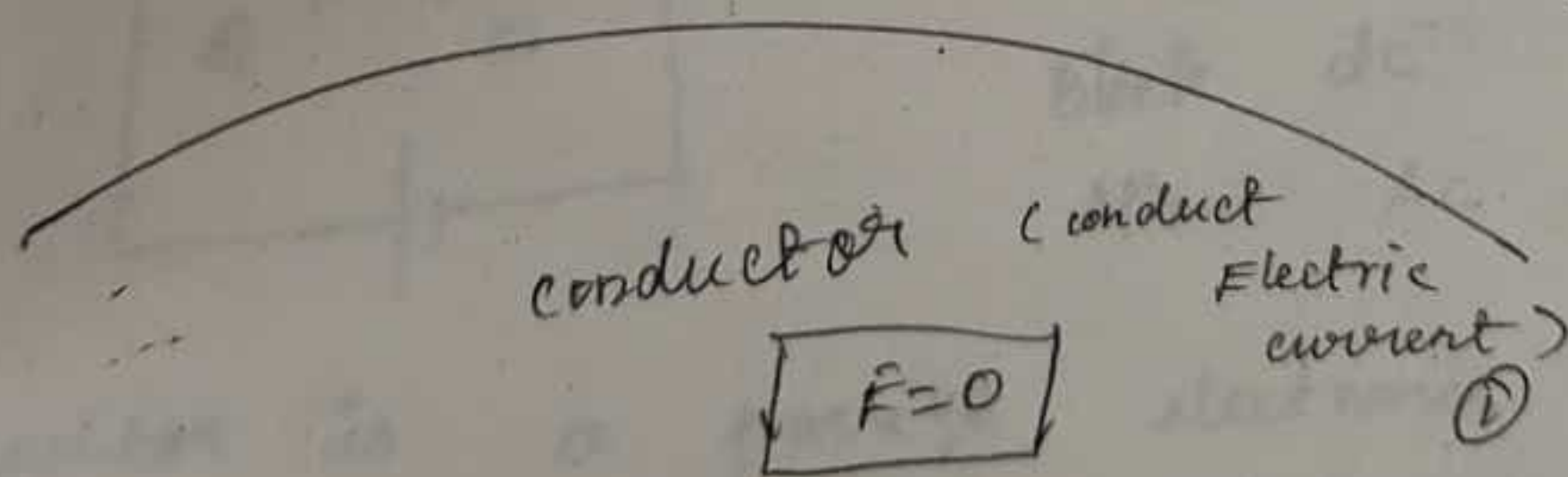
$$\Rightarrow \frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

Law of Refraction

conductor - dielectric boundary conditions: -

dielectric (ϵ_r)

②



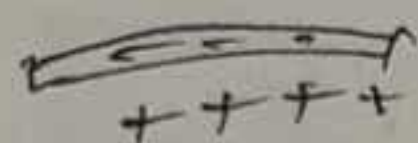
under static conditions: $\sigma \uparrow$

* Electric flux Normal direction.

i) $E = 0$ Inside the conductor.

ii) At the surface, $D_t = 0$ & $D_n = P_S$

* If the charges are enclosed by the box,
 $D_{n1} - D_{n2} = \rho_s$; $\rho_s \neq 0$

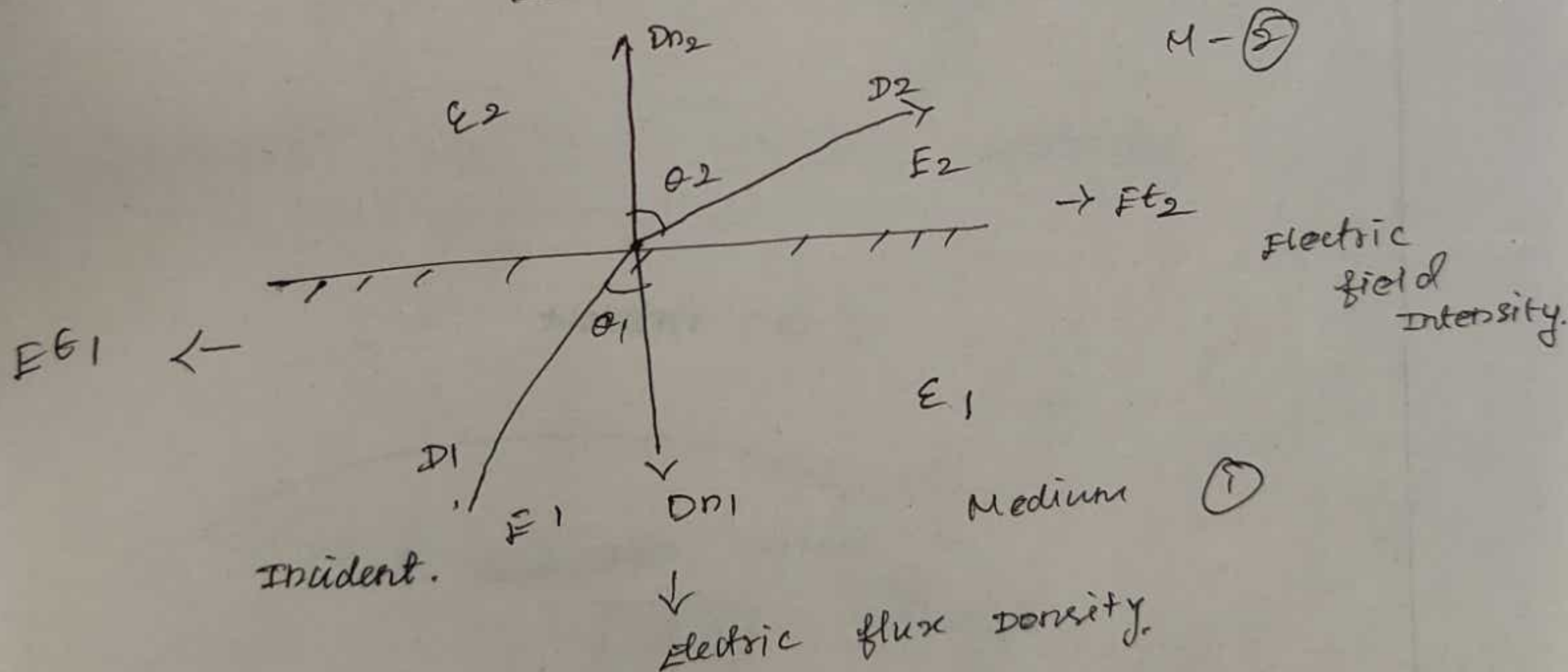
 ρ_s (charge).

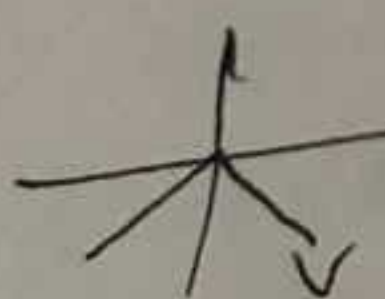
i.e) The electric flux density is discontinuous across the Boundary if $\rho_s \neq 0$.
Boundary conditions:

$$\left[\begin{array}{l} E_{t1} = E_{t2} \\ D_{n1} = D_{n2} ; \rho_s = 0 \\ D_{n1} - D_{n2} = \rho_s ; \rho_s \neq 0. \end{array} \right] \begin{array}{l} \text{---} \\ \text{---} \\ \text{---} \end{array}$$

Boundary conditions:

In terms of reflection & refraction at the Boundary.



 Reflection.

 transmission.

 Different Angle Refraction.

Law of Refraction b/w 2 dielectric Media.

* The normal component of 'D':

$$D_{n1} = D_1 \cos \theta_1$$

$$D_{n2} = D_2 \cos \theta_2$$

$$\Rightarrow 5V \quad \boxed{\text{---}}$$

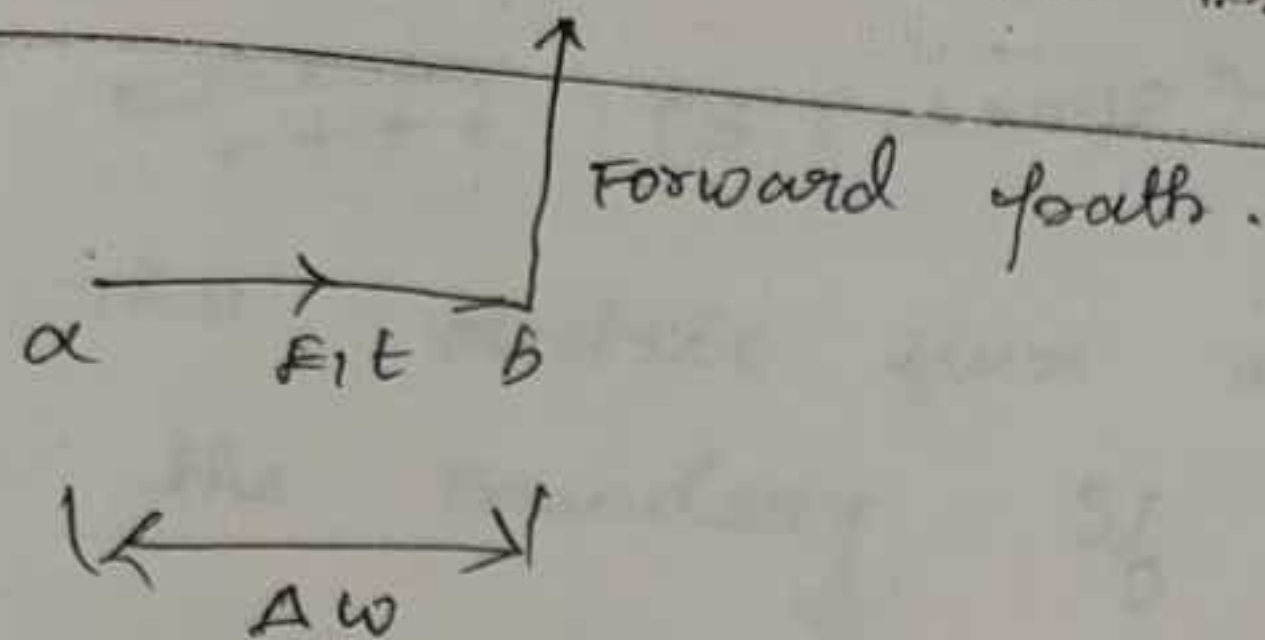
Total voltage is 0.

$$V_{\text{rise}} = V_{\text{drop}}$$

↓

கூடுதல் மீட்டர்கள்

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$$E_1 = E_{1t} + E_{1n}$$

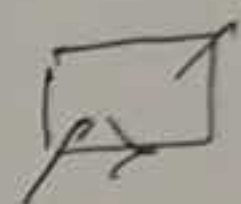
$$E_2 = E_{2t} + E_{2n}$$

In electrostatic field, the voltage around any closed path must be zero.

$$V = - \int \vec{E} \cdot d\vec{l} = 0$$

$$V = - \int \vec{E} \cdot d\vec{l}$$

Apply this in given Boundary Region.



$$E_{1t} \Delta w - E_{1n} \frac{\Delta h}{2} - E_{2n} \frac{\Delta h}{2} - \left[E_{2t} \Delta w - E_{2n} \frac{\Delta h}{2} - E_{1n} \frac{\Delta h}{2} \right]$$

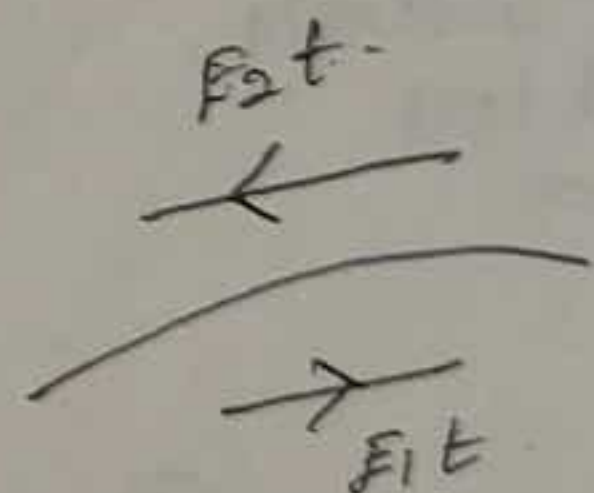
$$E_{1t} \Delta w - E_{1n} \frac{\Delta h}{2} - E_{2n} \frac{\Delta h}{2} - E_{2t} \Delta w + E_{2n} \frac{\Delta h}{2} + E_{1n} \frac{\Delta h}{2} \Rightarrow 0$$

$$E_{1t} \Delta w - E_{2t} \Delta w = 0$$

$$(E_{1t} - E_{2t}) \Delta w = 0$$

$$E_{1t} - E_{2t} = 0$$

$$\boxed{E_{1t} = E_{2t}}$$



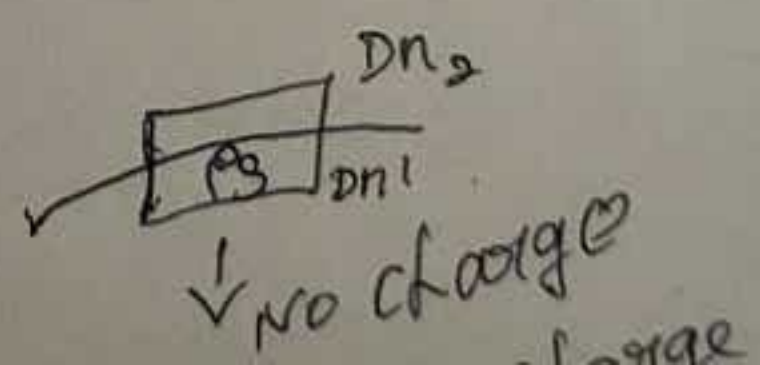
Thus, the tangential components of $\vec{E}$ are same on the 2 sides of the boundary.

E_t is continuous across the boundary.

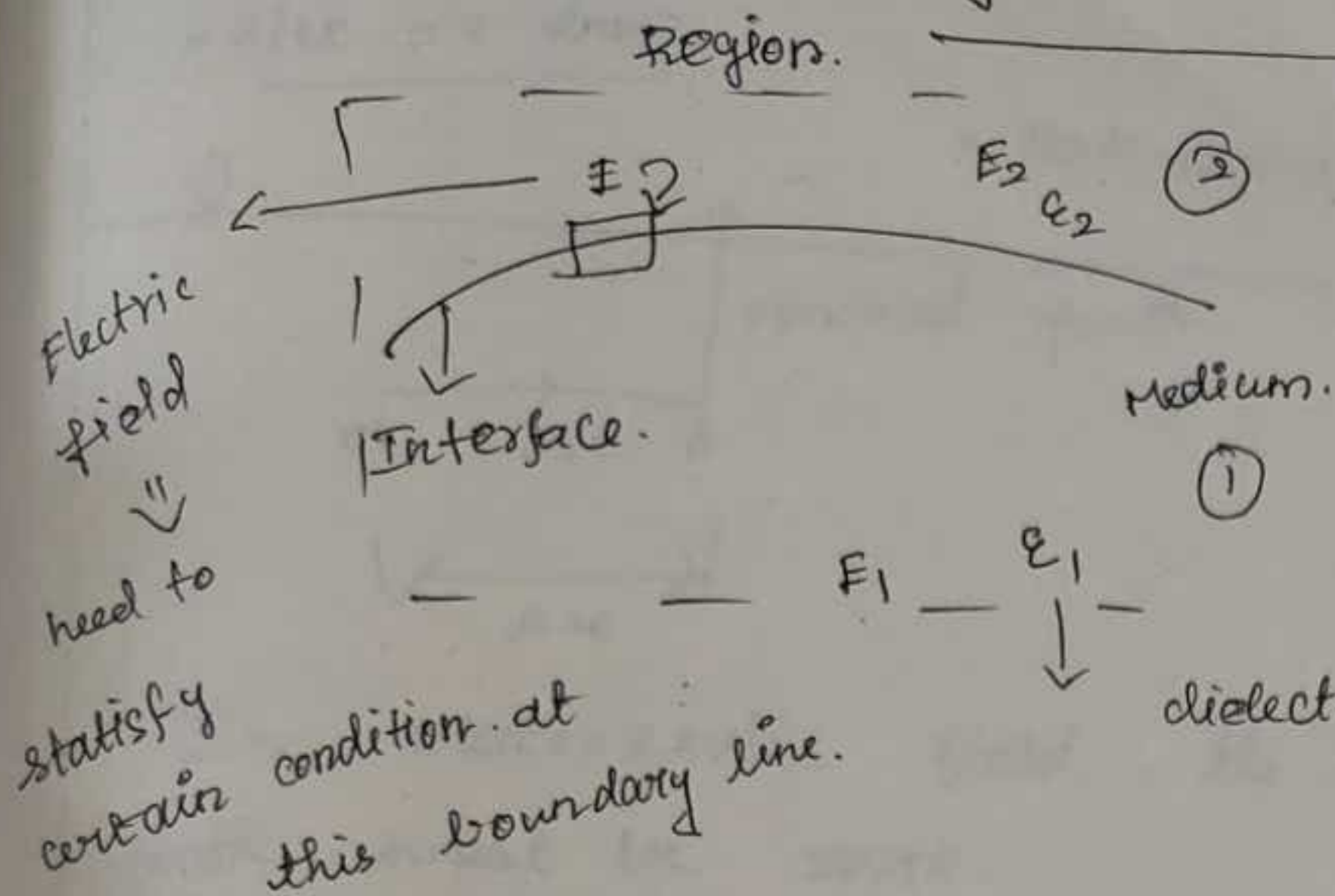
Electric flux density characteristics:

* If there are no charges enclosed by the box, (i.e) $\rho_s = 0$.

$$D_{n1} = D_{n2} ; \rho_s = 0$$



Boundary conditions for Electric field.



Interface between 2 different Medium.

→ If $\vec{E}$ exists in a region consisting of 2 different media, $\vec{E}$ must satisfy certain conditions at the interface (or) boundary.

→ Boundary conditions are used to determine the field characteristics on one side, if the field on the other side is known.

Types of Boundary:

i) Dielectric (ϵ_{r1}) - Dielectric (ϵ_{r2})

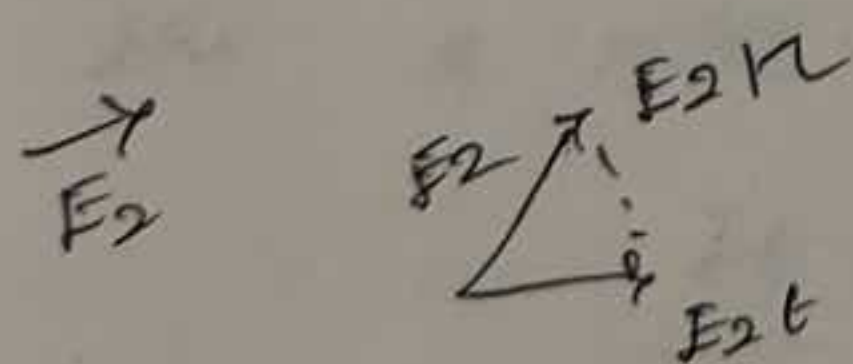
ii) conductor - Dielectric Boundary

iii) conductor - free space Boundary

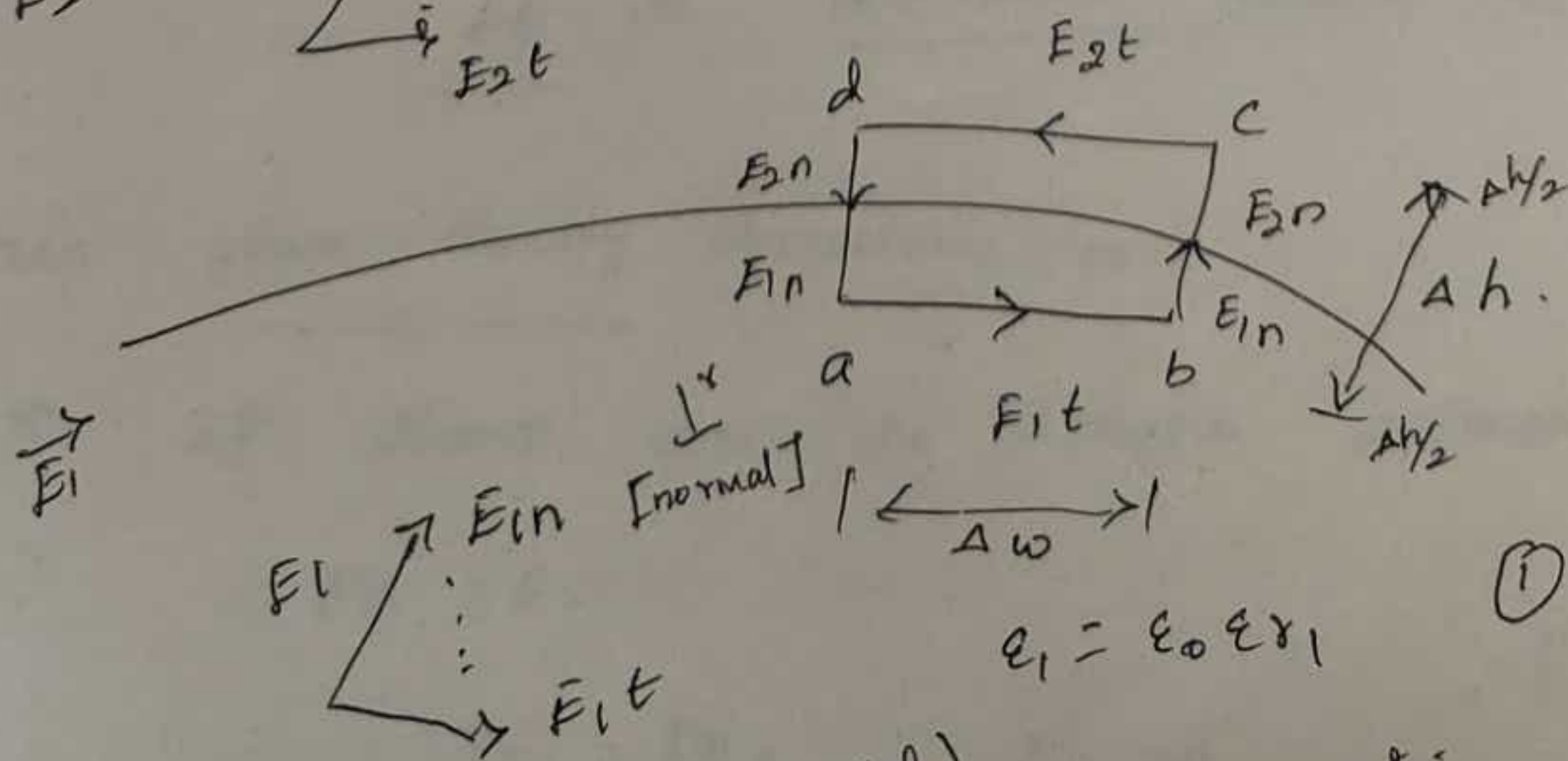
ϵ_{r2} wood.
 ϵ_{r1} (rubber)

Almost same.
→ $\epsilon_r = 1$

i) Dielectric - Dielectric Boundary conditions:



$\epsilon_2 = \epsilon_0 \epsilon_{r2}$ permittivity.
that is dielectric constant.



$$\vec{E}_1 = \vec{E}_{1t} + \vec{E}_{1n}$$

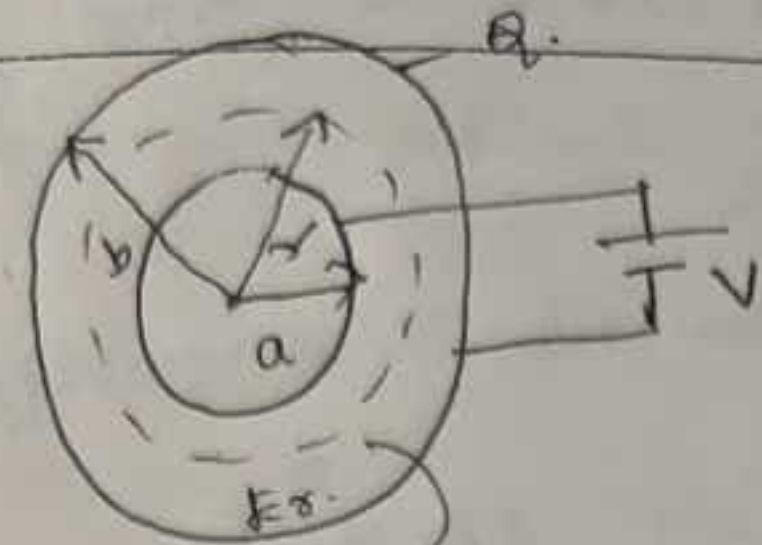
total (tangential) // to component.

① Electric field $\vec{E}$ intensity

② Electric flux density $\vec{D}$

of the conductor is it contain.

capacitance of concentric sphere:



Energy store

→ consider two concentric spheres of inner radius 'a' and outer radius 'b'

→ the electric field intensity at any point in b/w inner & outer sphere is.

$$E = \frac{Q}{4\pi\epsilon r^2} ; a \leq r \leq b.$$

→ The potential difference between the sphere.

$$V = - \int_b^a \frac{Q}{4\pi\epsilon r^2} dr = - \frac{Q}{4\pi\epsilon} \int_b^a \frac{1}{r^2} dr \quad | \quad V = - \int E \cdot dl$$

$$= - \frac{Q}{4\pi\epsilon} (-1) r^{-1} \Big|_b^a \Rightarrow \frac{Q}{4\pi\epsilon} \left[\frac{1}{a} - \frac{1}{b} \right]$$

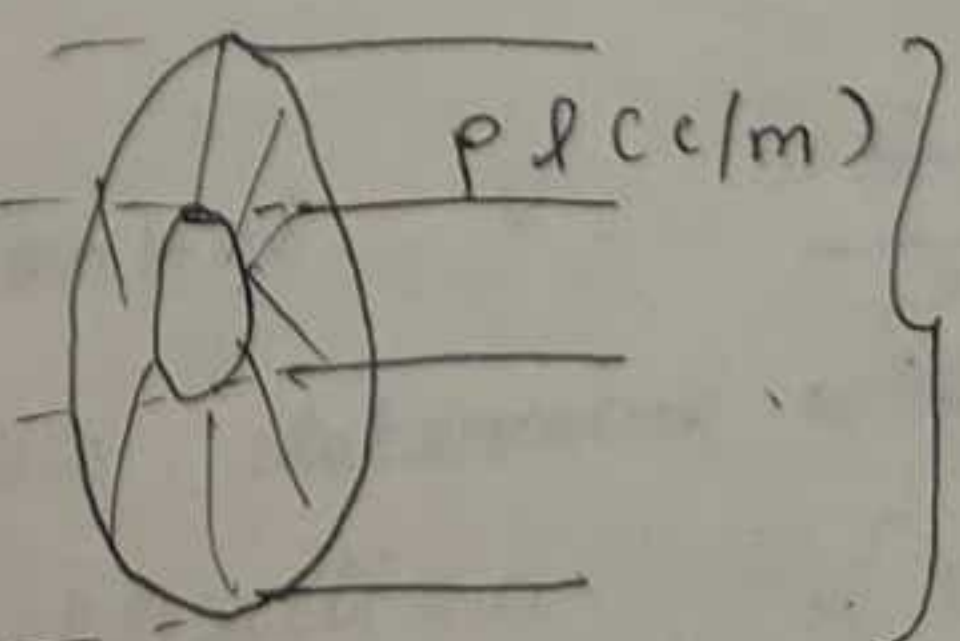
$$V = \frac{Q}{4\pi\epsilon} \left[\frac{b-a}{ab} \right] \Rightarrow \frac{Q}{V} = \frac{4\pi\epsilon (ab)}{b-a}$$

$$C = 4\pi\epsilon \left[\frac{ab}{b-a} \right] ; \text{Farad.}$$

$$C = \frac{Q}{V}$$

capacitance of coaxial cable [cylindrical]

$\frac{F}{L}$



Two cylinders are charged at the rate of 'PL'

$$PL = \frac{Q}{l}$$

Charge per length.

of the conductor is directly contain.

for sphere coordinate system:

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 V}{\partial \phi^2} = -\rho V / \epsilon$$

The flux density:

$$D_1 = \frac{Q}{A} \Rightarrow \epsilon_1 E_1 = \frac{Q}{A}$$

$$D_1 \Rightarrow E_1 = \frac{Q}{A \epsilon_0 \epsilon r_1}$$

$$D_2 = E_2 = \frac{Q}{A \epsilon_0 \epsilon r_2} \rightarrow \text{substitute in (1).}$$

$$V = \frac{Q}{A \epsilon_0 \epsilon r_1} d_1 + \frac{Q}{A \epsilon_0 \epsilon r_2} d_2$$

$$V = \frac{Q}{A \epsilon_0} \left[\frac{d_1}{\epsilon r_1} + \frac{d_2}{\epsilon r_2} \right]$$

$$V = \frac{Q}{A \epsilon_0} \left[\frac{d_1 \epsilon r_2 + d_2 \epsilon r_1}{\epsilon r_1 \epsilon r_2} \right] \Rightarrow$$

$$C = \frac{Q}{V} \Rightarrow \frac{A \epsilon_0 [\epsilon r_1 \epsilon r_2]}{d_1 \epsilon r_2 + d_2 \epsilon r_1} ; \text{Farad.}$$

dielectric media also there.

capacitance of an isolated sphere:

(spherical.)

* taking radius as line integration so dr.



Q.

→ consider a sphere of radius 'r' having a charge Q.

→ the electric potential is given as.

$$V = - \int E \cdot dr ; E = \frac{Q}{4\pi \epsilon r^2} \hat{r} \Rightarrow \text{Electric field intensity. } V = E \cdot d$$

$$V = - \int \frac{Q}{4\pi \epsilon r^2} \cdot dr$$

instead of taking distance.

$$K \frac{1}{d} V = E \cdot d$$

from Gauss law.

$$= - \frac{Q}{4\pi \epsilon} \int \frac{1}{r^2} \cdot dr \Rightarrow - \frac{Q}{4\pi \epsilon} \int r^{-2} \cdot dr$$

$$= - \frac{Q}{4\pi \epsilon} \left[(-2) r^{-1} \right]$$

$$\Rightarrow \int r^{-2} = -2$$

$$V = \frac{Q}{4\pi \epsilon r} \Rightarrow C = \frac{Q}{V} = 4\pi \epsilon r ;$$

$$\int r^{-2} = (-1) r^{-2+1} = -1 r^{-1}$$

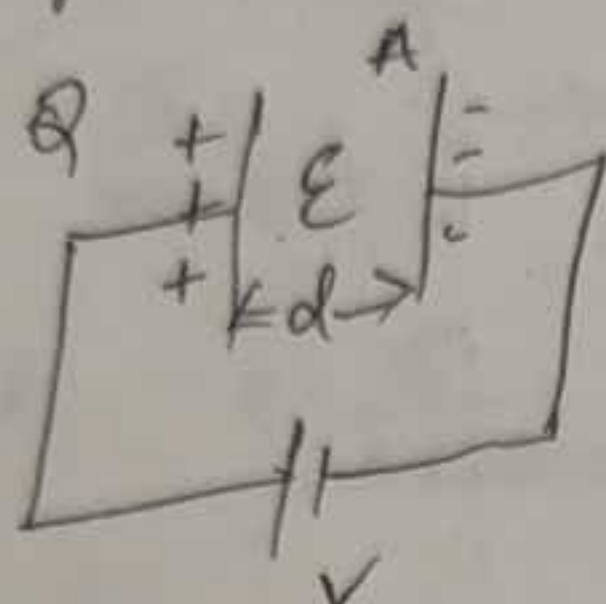
Farad.

* Derive the capacitance of parallel capacitors, spherical capacitors, coaxial and transmission line.

→ capacitance of கூடுதல் மீட்டர்

↓ cylindrical.

11th plate capacitor.



consider a capacitor composed of 2 conducting plates of area 'A'

$\epsilon \rightarrow$ dielectric constant.

$\rightarrow A$ uniform electric flux density over the plates.

$$D = Q/A ; \text{ C/m}^2$$

$$[D = \epsilon E]$$

$$\epsilon E = Q/A \Rightarrow Q = \epsilon E A \quad \text{--- (1)}$$

Relation b/w E & V .

$$E = -\nabla V$$

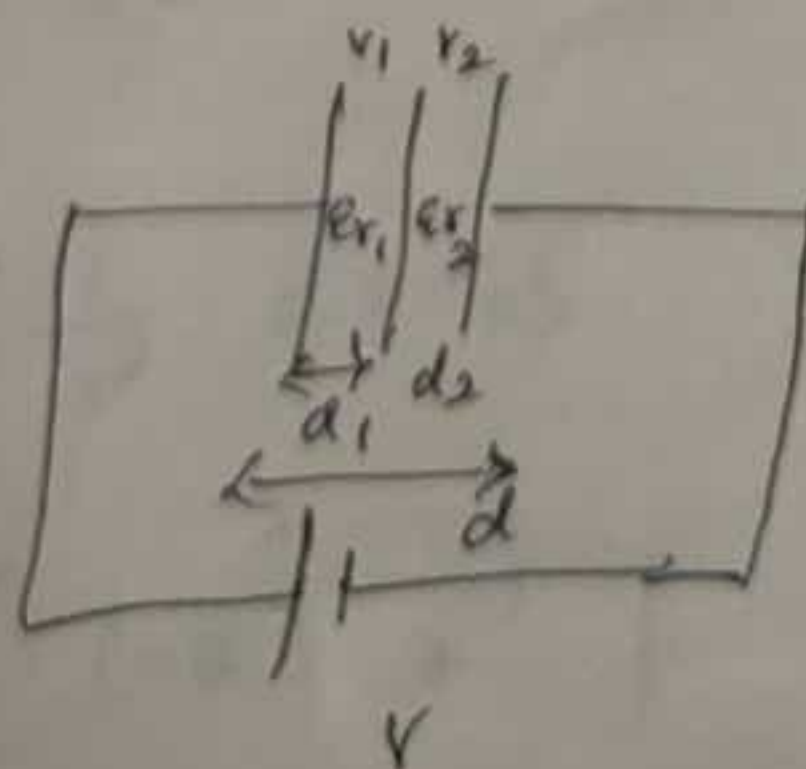
$$V = -\int E dl \Rightarrow \int E dl \Rightarrow E \int dl = E \cdot d$$

$$V = E \cdot d \Rightarrow E = V/d \rightarrow \text{substitute in (1)}$$

$$Q = \frac{\epsilon A V}{d} \Rightarrow \frac{Q}{V} = \frac{\epsilon A}{d}$$

$$\therefore C = \frac{\epsilon A}{d} ; \text{ Farad. } \hat{X}$$

* capacitance of 2 dielectric media:-



* consider a 11th plate capacitor with 2 dielectric media $[\epsilon_1 \text{ \& } \epsilon_2]$.

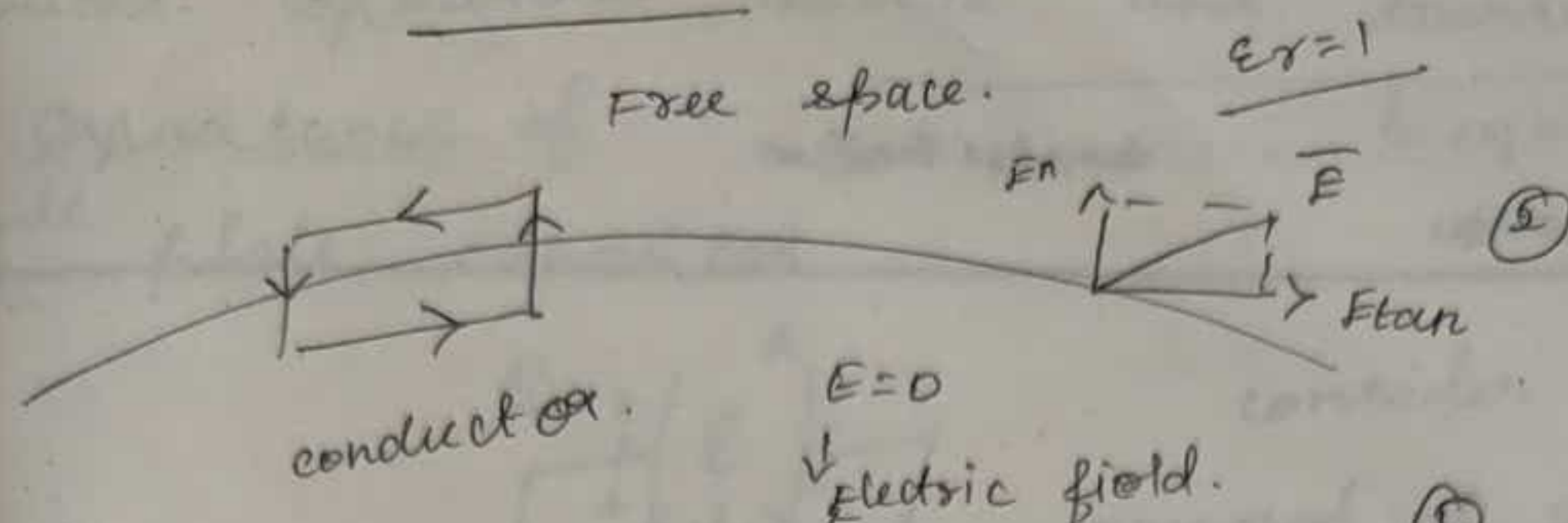
$\rightarrow$ the applied potential V is

$$V = V_1 + V_2$$

$$V = E_1 d_1 + E_2 d_2$$

--- (1)

Conductor - Free space Boundary conditions:



→ The tangential component of $\vec{E}$ is zero at the boundary.

$$E_t = 0$$

$D_t = 0$

$$D_t = \epsilon E_t$$

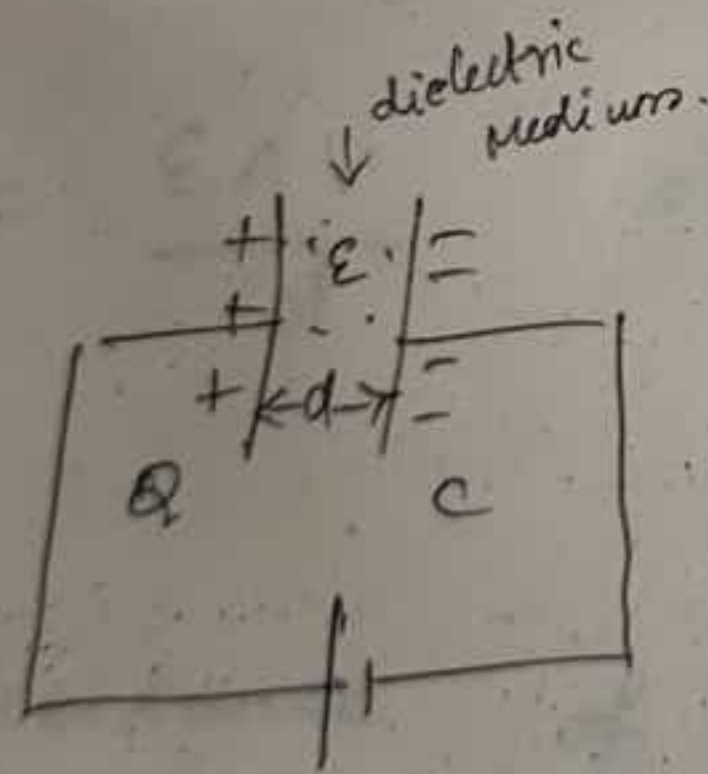
→ Only D_n & E_n exist at the boundary (i.e.) always in perpendicular direction.

$$D_n = \rho_s$$

$$E_n = \rho_s / \epsilon_0 \quad ; \quad \epsilon_r = 1 \text{ for free space.}$$

Capacitor of capacitance.

Capacitor: [2 terminal electronic passive device to store Electric Energy]



$$C = \frac{Q}{V} \quad ; \quad C = \frac{A\epsilon}{d}$$

Block dc

Allow Ac.

⇒ A capacitor is a passive electronic component with two terminals to store electric energy.

→ It blocks the flow of DC and permits the AC.

→ property of a capacitor is known as capacitance [C]

unit → Farad.

A → cross sectional area

d → distance b/w the plates.

the conductor is directly to the charge it contains.

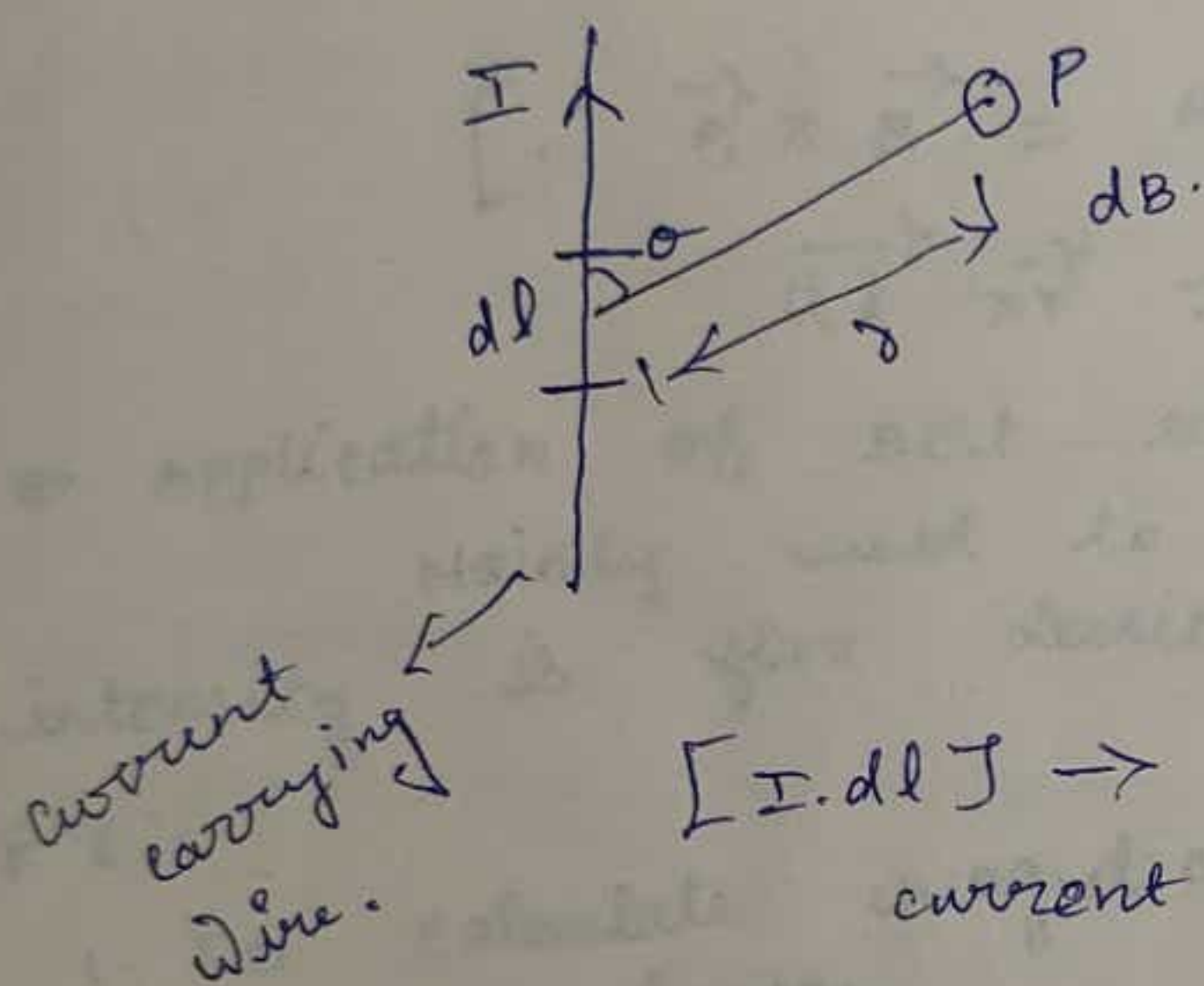
for sphere coordinate system
 $\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial V}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial V}{\partial \theta}) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 V}{\partial \phi^2} = -\rho_v / \epsilon_0$
 $\rightarrow$ must upto.

$\frac{1}{4\pi} \frac{dm(\theta)}{bm}$

Unit-3

Biot - Savart law: [To relate magnetic field and electric current]

The magnetic flux density produced by a current element at any point in a magnetic field is,
 $\rightarrow$ proportional to the current element. ($I dl$)
 $\rightarrow$ inversely proportional to the square of the distance between them (r^2).



current carrying conductor I produce Magnetic field around it

In magnetic field
 $\downarrow$
 (Magnetic flux) at that point.

[$I \cdot dl$] $\rightarrow$ current element $\uparrow$ dB $\uparrow$ point.

Flux density.

$$dB \propto I dl$$

$$dB \propto \frac{1}{r^2}$$

$$dB \propto \sin \theta$$

It can be written as..

$$dB = \frac{\mu}{4\pi} \times \frac{I dl \sin \theta}{r^2}$$

$$dB \propto \frac{I dl \sin \theta}{r^2}$$

$$dB = \frac{\mu}{4\pi} \times \frac{I dl \sin \theta}{r^2}$$

current
 length
 distance
 direction

where,

$I dl \rightarrow$ current element.

$r \rightarrow$ The distance between $I dl$ & the point 'P'

$\mu \Rightarrow \mu_0 \mu_r \rightarrow$ permeability.

the potential of the conductor is directly proportional to the charge it contains.

for sphere coordinate system:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \Rightarrow \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = -\frac{\rho}{\epsilon_0}$$

$$\frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = -\frac{\rho r^2}{\epsilon_0}$$

Electrostatic boundary-value problems.

capacitance of a coaxial cable [cylindrical capacitor].
கருத்து பற்றி

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11/5/2021

$$C = \frac{2\pi\epsilon l}{\ln(b/a)}; \text{ F/m } [l=1 \text{ length of a cable}]$$

capacitance of an isolated sphere.

capacitance of concentric sphere.

$$C = 4\pi\epsilon r; \text{ Farad.}$$

$$C = 4\pi\epsilon \left[\frac{ab}{b-a} \right]; \text{ F.}$$

Pb.

① A parallel plate capacitor has an area of 0.8 cm^2 separation of 0.1 mm with a dielectric for which $\epsilon_r = 1000$ is a field of 10^6 V/m . calculate C & V .

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$V = E \cdot d$$

$$C = 70.83 \text{ nF}$$

$$V = 100 \text{ V.}$$

② Evaluate the capacitance of
i) A spherical satellite 1.5 m diameter in free space.

$$C = 4\pi\epsilon r.$$

ii) A coaxial cable 1.5 m long filled with polyethylene ($\epsilon_r = 2.26$) with inner conductor of radius 0.6 mm and inner radius of outer conductor 3.5 mm .

$$C = \frac{2\pi\epsilon l}{\ln(b/a)}$$

Assignment.

⇒ laws & equations:

1. coulomb's law

2. problems:

1. Find the force between 2 charges spaced 10cm apart in vacuum. The charges are $4 \times 10^{-8} \text{C}$ and $6 \times 10^{-5} \text{C}$ respectively. If the same charges are separated by the same distance in kerosene with $\epsilon_r = 2$, what is the force of attraction?

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} ; F = \frac{Q_1 Q_2}{4\pi\epsilon_0 \epsilon_r R^2} \quad [1.0785 \text{N}]$$

$$[\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m} ; \epsilon_r = 1 ; \epsilon_r = 2]$$

2. Find the force in Newton b/w 2 charges $30 \mu\text{C}$ of $-30 \mu\text{C}$ situated at $(0, 1, 2) \text{ m}$ and $(2, 0, 0) \text{ m}$ resp. in free space. $[0.9 \text{ N}]$

$$\Rightarrow r = \sqrt{(2-0)^2 + (0-1)^2 + (0-2)^2} = 3$$

$$q_1, q_2 = \epsilon_r = 1 \quad \therefore \epsilon_0 =$$

Gauss law.

pbm. $\psi = \phi = \oint_S \rho \cdot d\mathbf{s} ; \phi = \frac{\phi}{\epsilon} = \oint_S \mathbf{F} \cdot d\mathbf{s}.$

1. If $\vec{D} = xxy\vec{a}_x + 3yz\vec{a}_y + 4zx\vec{a}_z$, find the charge enclosed by $-1 \leq x \leq 2$; $0 \leq z \leq 4$ & $y=3$ plane.

sol:

Poisson's equation

$$\nabla^2 V = -\frac{\rho V}{\epsilon} \quad \nabla^2 V = 0.$$

Capacitance:

$$C = \frac{Q}{V}$$

series

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

||le.

$$C_{eq} = C_1 + C_2 + \dots + C_n$$

→ consider a coaxial cable of inner radius 'a' & outer radius 'b'.
By applying Gauss's law, $\vec{E}$ at any distance 'r' between a and b.

$$\vec{E} = \frac{\rho l}{2\pi\epsilon r} \hat{r} \quad \text{cylindrical.}$$

potential difference between 2 cylinders:

$$V = - \int_b^a \vec{E} \cdot d\vec{r} \Rightarrow - \int_b^a \frac{\rho l}{2\pi\epsilon r} dr \quad \left| \begin{array}{l} E = \frac{Q}{4\pi\epsilon r^2} \end{array} \right.$$

$$= - \frac{\rho l}{2\pi\epsilon} \int_b^a \frac{1}{r} dr \Rightarrow - \frac{\rho l}{2\pi\epsilon} (\ln r)_b^a$$

(Natural logarithm of r.)

$$\Rightarrow - \frac{\rho l}{2\pi\epsilon} [\ln a - \ln b] \Rightarrow \frac{\rho l}{2\pi\epsilon} [\ln b - \ln a]$$

$$V = \frac{\rho l}{2\pi\epsilon} \ln(b/a)$$

$$C = \frac{Q}{V} = \frac{\rho l}{V}$$

capacitance of coaxial cable / unit.

$$C = \frac{\rho l}{V} \Rightarrow C = \frac{2\pi\epsilon}{\ln(b/a)} \quad ; \text{ F/m.}$$

uniqueness theorem.

If the solution of Laplace's equation $\nabla^2 V = 0$ satisfies the boundary condition, then the solution is unique, by whatever method it is used.

problem:

Given the potential $V = \frac{50 \sin \theta}{r^2}$

in free space, determine whether

V satisfies Laplace equation. $[\nabla^2 V = 0]$

Ans $\nabla^2 V \neq 0$

The direction of the force is same as the direction of $\vec{E}$.

$F_m \rightarrow$ magnetic force: The force experienced by a charged particle moving with velocity (v) in a magnetic field of ' B '

$$\vec{F}_m = Q (\vec{v} \times \vec{B})$$

substitute (2) & (3) in (1) — (3)

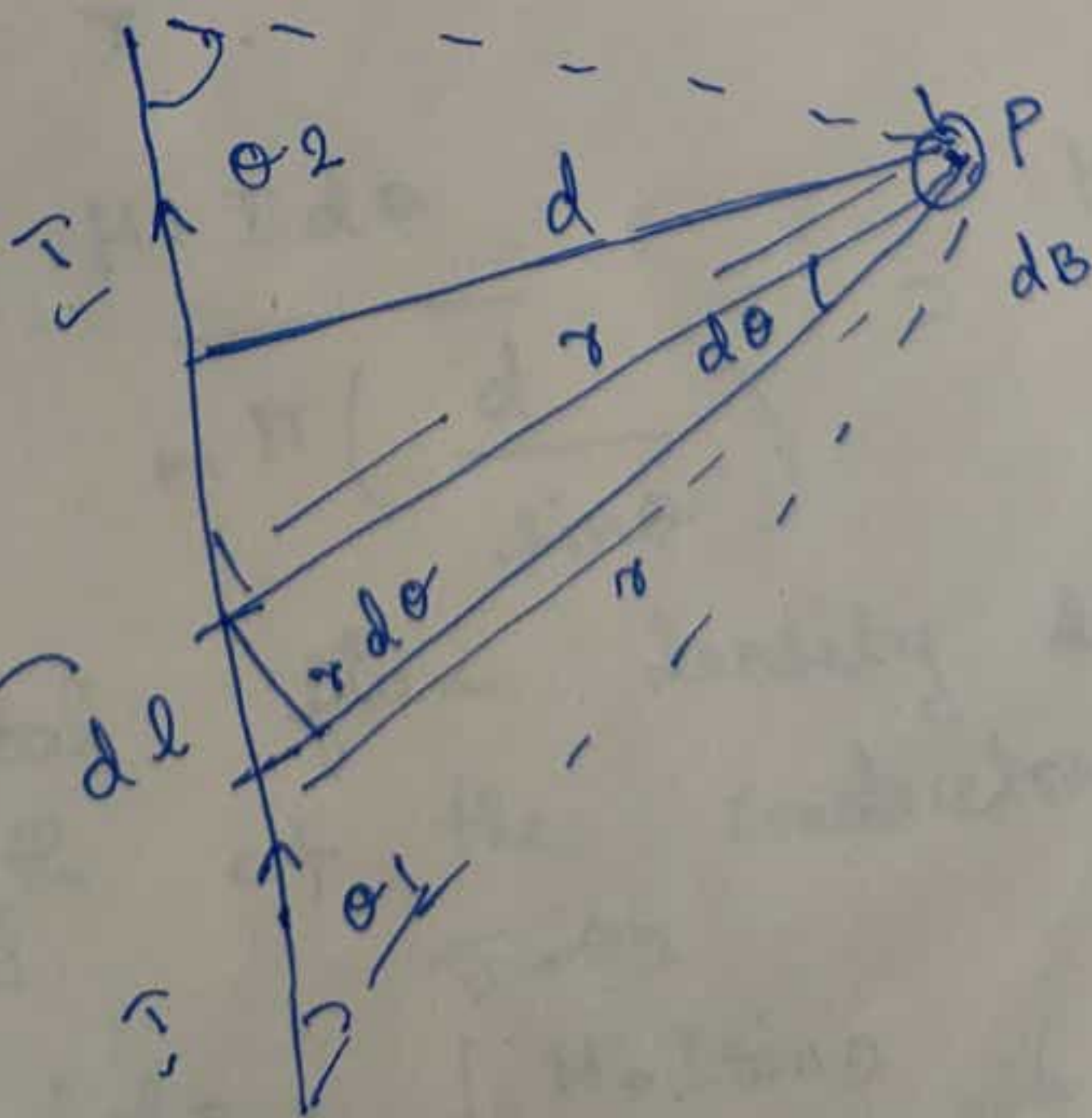
$$\vec{F} = Q\vec{E} + Q(\vec{v} \times \vec{B})$$

$$\vec{F} = Q [\vec{E} + (\vec{v} \times \vec{B})]$$

This equation is called as Lorentz force equation (2m & 8m).

* Magnetic field intensity (H) due to finite & infinite conductor. (or) Derive the Application of Biot-Savart law.

$\rightarrow$ consider a conductor of finite length with a current ' I ' flowing through it.

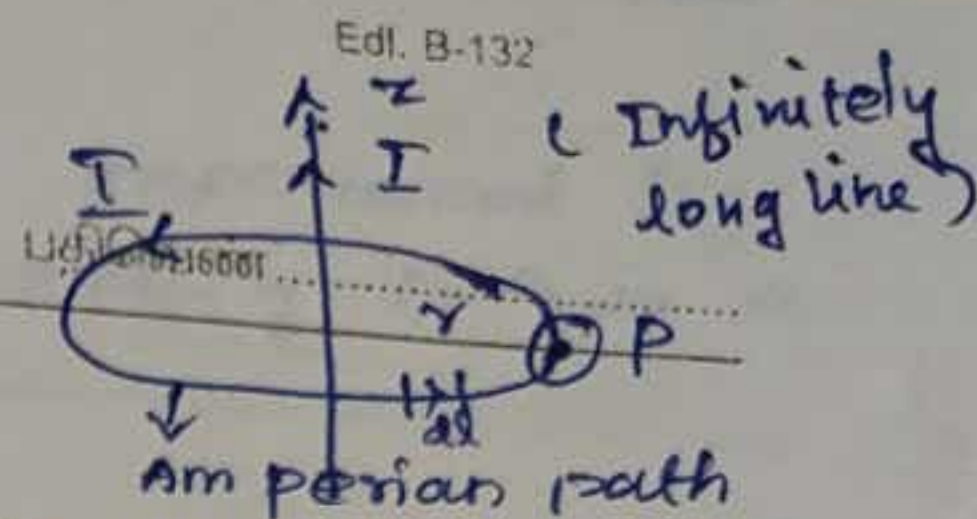


(current differential length element)

$\rightarrow$ consider a small current element ' $I dl$ ' in the conductor.

$\vec{H}$ of an infinitely long line carrying conductor.

→ consider an infinitely long line which is carrying the current 'I'
 → let 'P' be any point at which $\vec{H}$ has to be found.
 → consider a closed path passing through the point 'P' (i.e.) Amperian path.



According to Ampere's law,

$$\oint \vec{H} \cdot d\vec{l} = I$$

$$\vec{H} \oint dl = I$$

$$\vec{H} (2\pi r) = I$$

$$\vec{H} = \frac{I}{2\pi r} \vec{a}_\phi ; \text{ A/m.}$$

$$[\because \oint dl = 2\pi r]$$

↓
circumference of the circle.

↓
direction of Magnetic field.

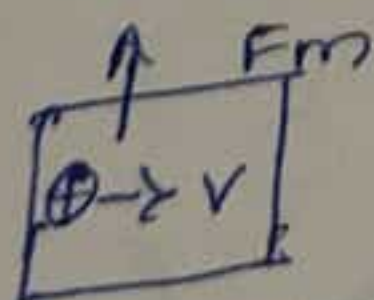
Lorentz force Equation. (Electric Motors)

The force on a moving particle due to combined electric and magnetic fields is given by,

$$\vec{F} = \vec{F}_E + \vec{F}_m \quad (1)$$

Electric field (experience force)
 $\vec{F}_E = Q\vec{E}$

moving charge placed in magnetic field



(Magnetic force).

$\vec{B}$ flux density.

$$F_m \propto B.$$

where,

$\vec{F}_E$ → electric force: The force acting on a charged particle.

$$\vec{F}_E = Q\vec{E}$$

— (2)

of the conductor as
the charge it contain.

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \phi} \left(\sin \theta \frac{\partial V}{\partial \phi} \right) = -\rho / \epsilon_0$$

→ more upto.

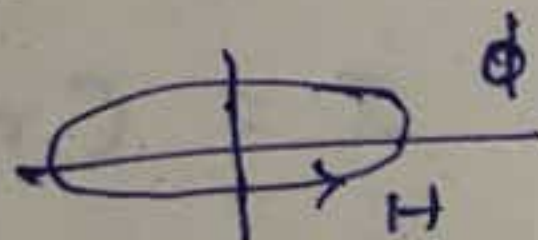
Applications of Ampere's law.

Ampere's law is widely used to calculate the magnetic fields of current distributions with a high degree of symmetry.

Ex: → To measure the magnetic field intensity of an infinite line current. $\vec{H}$ of an infinite long coaxial transmission line.

Amperean path:

→ The closed loop to which Ampere's law is applied → Amperean loop.



closed path is known as Amperean path.

Ampere's law in point form.

→ According to Ampere's law, $\downarrow$ differential form.

$$\oint \vec{H} \cdot d\vec{l} = I \quad (1)$$

From the relation b/w I & J , [current & current density J].

$$I = \iint_S \vec{J} \cdot d\vec{S} \quad (2)$$

comparing (1) & (2)

$$\oint_L \vec{H} \cdot d\vec{l} = \iint_S \vec{J} \cdot d\vec{S} \quad (3)$$

line & surface integral

Applying Stoke's theorem.

$$\oint_L \vec{H} \cdot d\vec{l} = \iint_S (\nabla \times \vec{H}) \cdot d\vec{S} \quad (4)$$

From (3) & (4)

$$\iint_S (\nabla \times \vec{H}) \cdot d\vec{S} = \iint_S \vec{J} \cdot d\vec{S}$$

$$\boxed{\nabla \times \vec{H} = \vec{J}} \quad \text{A/m}^2$$

Biot - Savart law in vector form:

$$d\vec{B} = \frac{\mu}{4\pi} \cdot \frac{I d\vec{l} \sin\theta}{r^2} \times \frac{\vec{r}}{r}$$

கூடுதல் புத்தகம்

Ex B-132

$$= \frac{\mu}{4\pi} \cdot \frac{I d\vec{l} \cdot \vec{r} \sin\theta}{(r^2)^{3/2}}$$

$$d\vec{B} = \frac{\mu}{4\pi} \cdot \frac{I (d\vec{l} \times \vec{r})}{r^3}$$

$$[\because \vec{A} \times \vec{B} = AB \sin\theta]$$

$$d\vec{l} \times \vec{r} = d\vec{l} r \sin\theta$$

* Application of Biot - Savart law:

Mainly used to measure the magnetic field intensity & flux density at various points.

Ex:-

1. TO calculate magnetic field due to finite and infinite conductor.
2. Magnetic field intensity on the axis of circular coil.
3. " " " " Rectangular coil.
4. " " " " Magnetic field.

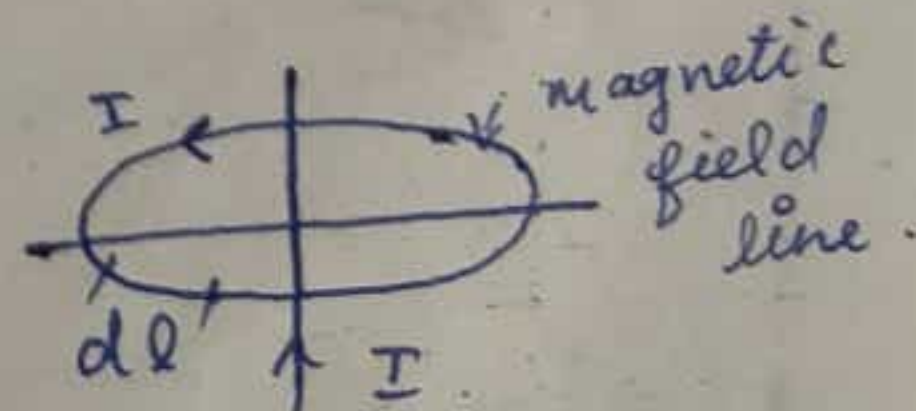
Ampere's law. -X- (short cut)

Ampere's law relates the net magnetic field along a closed loop to the electric current passing through the loop.

Statement:

$$\oint \vec{H} \cdot d\vec{l} = I$$

The line integral of magnetic field intensity 'H' around any closed path is equal to the total current 'I' enclosed by that path. -X-



⊕ Rest Electric field.
Moving → Magnetic field.

for sphere

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 V}{\partial \phi^2} = -\rho V / \epsilon_0$$

→ mark upto.

Laplace Equation:-

main condition $\nabla^2 V = 0$

have only equation

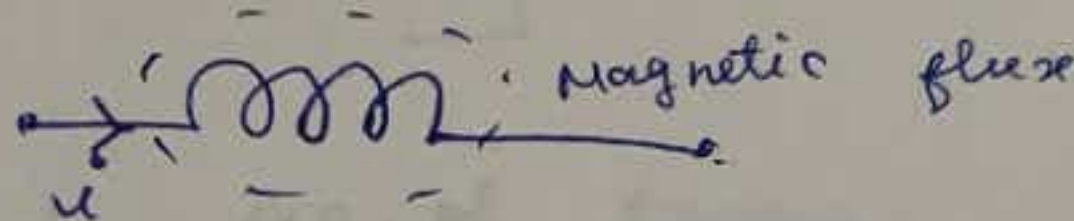
Inductor and Inductance super power.

Def:

An inductor is a passive two terminal electronic component.

It stores energy in magnetic field when electric current flows through it.

Also called as coil, choke (or) reactor.



It uses a conductor that is wound into a coil.

Application:

→ It is widely used in electrical and electronic devices for.

choking

Blocking

attenuating

filtering / smoothing high frequency noise.

storing and transferring energy in power converters.

* Inductance is the property of an Inductor.



Inductance is the property of an electric circuit by which an emf (voltage) is induced in it by a changing magnetic field.

An emf is induced to oppose the changing current.

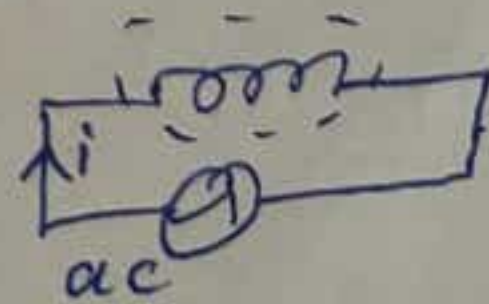
Types:

self inductance

(L)

Mutual Inductance

(M)



①

Inductance:

↳ capacitor in electric field.

likewise.

Inductance in Magnetic field. (Store) energy.

↳ Noise ↓.

proper voltage (power supply)

Eg from [phone charger to power converter.

Habit $\Rightarrow$ Inductance (other components do). [EMF].

(Any changes) (stable condition).

② voltage

current

Magnetic field.

$$\frac{I}{4\pi r} = H$$

If the conductor is infinitely long
(i.e) $\theta_1 = \theta_2 = 0$, then the flux density is given as,

From (3) $\Rightarrow B = \frac{\mu_0 I}{4\pi d} [\cos 0^\circ + \cos 0^\circ]$

$$B = \frac{\mu_0 I}{2\pi d} \quad (2)$$

$$= \frac{\mu_0 I}{2\pi d} ; \text{ Weber/m}^2$$

Then, the magnetic field intensity can be obtained as.

$$H = \frac{B}{\mu_0}$$

using
↓ relation.
 $B = \mu_0 H$

$$H =$$

$$H = \frac{\mu_0 I}{2\pi d} \cdot \frac{1}{\mu_0}$$

$$H = \frac{I}{2\pi d} ; \text{ A/m}$$

The total magnetic flux linkage is given as.

$$\phi = \int B \cdot dS.$$

↓ changing so using $\int$

where,

$B \rightarrow$ magnetic flux density.

$$B = \frac{\mu_0 N I}{l}$$

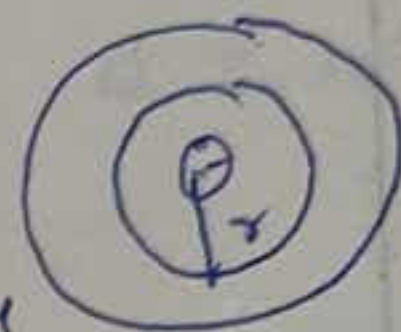
$$[\because N=1]$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$l = 2\pi r \text{ (circumference of circle)}$$

$$\phi = \int_a^b \frac{\mu_0 I}{2\pi r} \cdot dr$$

[inner conductor to outer conductor limit at a to b limit]



$$= \frac{\mu_0 I}{2\pi} \int_a^b \left(\frac{1}{r}\right) dr$$

Area $\rightarrow r \cdot dr$ (thickness)

$$= \frac{\mu_0 I}{2\pi} \left[\ln r \right]_a^b$$

$$\int \frac{1}{x} dx \Rightarrow \ln x$$

$$\phi = \frac{\mu_0 I}{2\pi} [\ln b - \ln a] = \frac{\mu_0 I}{2\pi} \ln(b/a)$$

$$\phi = \frac{\mu_0 I}{2\pi} \ln(b/a) \quad \text{--- (2) ---}$$

substit in (1)

$$L = \frac{\frac{\mu_0 I}{2\pi} \ln(b/a)}{I}$$

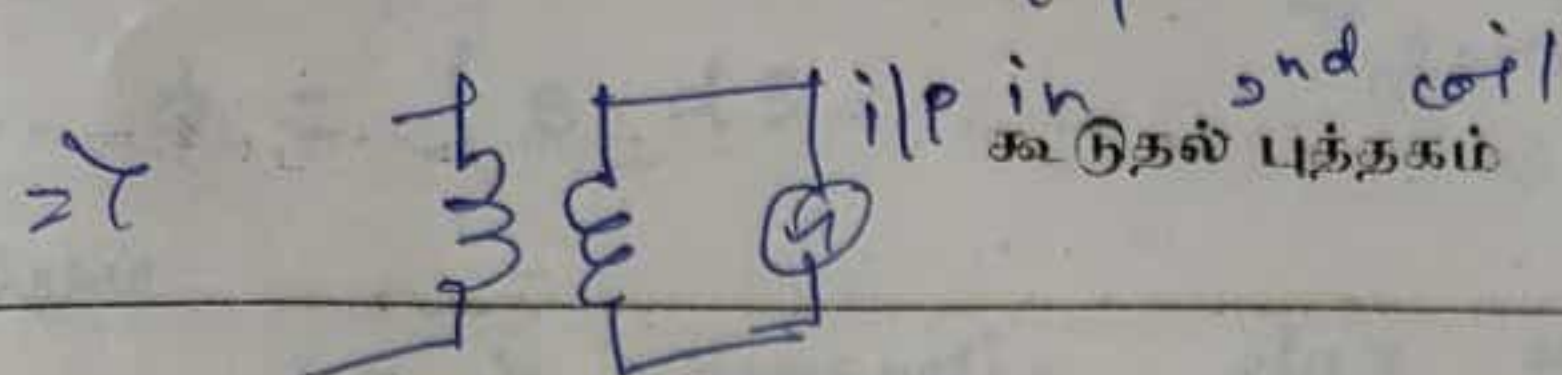
Non linear drop so log

$$L = \frac{\mu_0}{2\pi} \ln(b/a) \text{ H/m}$$

↓

$$M = \frac{N_2 \Phi_{12}}{i_1}$$

Edl. B-132



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coupling coefficient.

How much flux is produced in second coil.

$$K = \frac{\Phi_{12}}{\Phi_1} = \frac{\Phi_{21}}{\Phi_2}$$

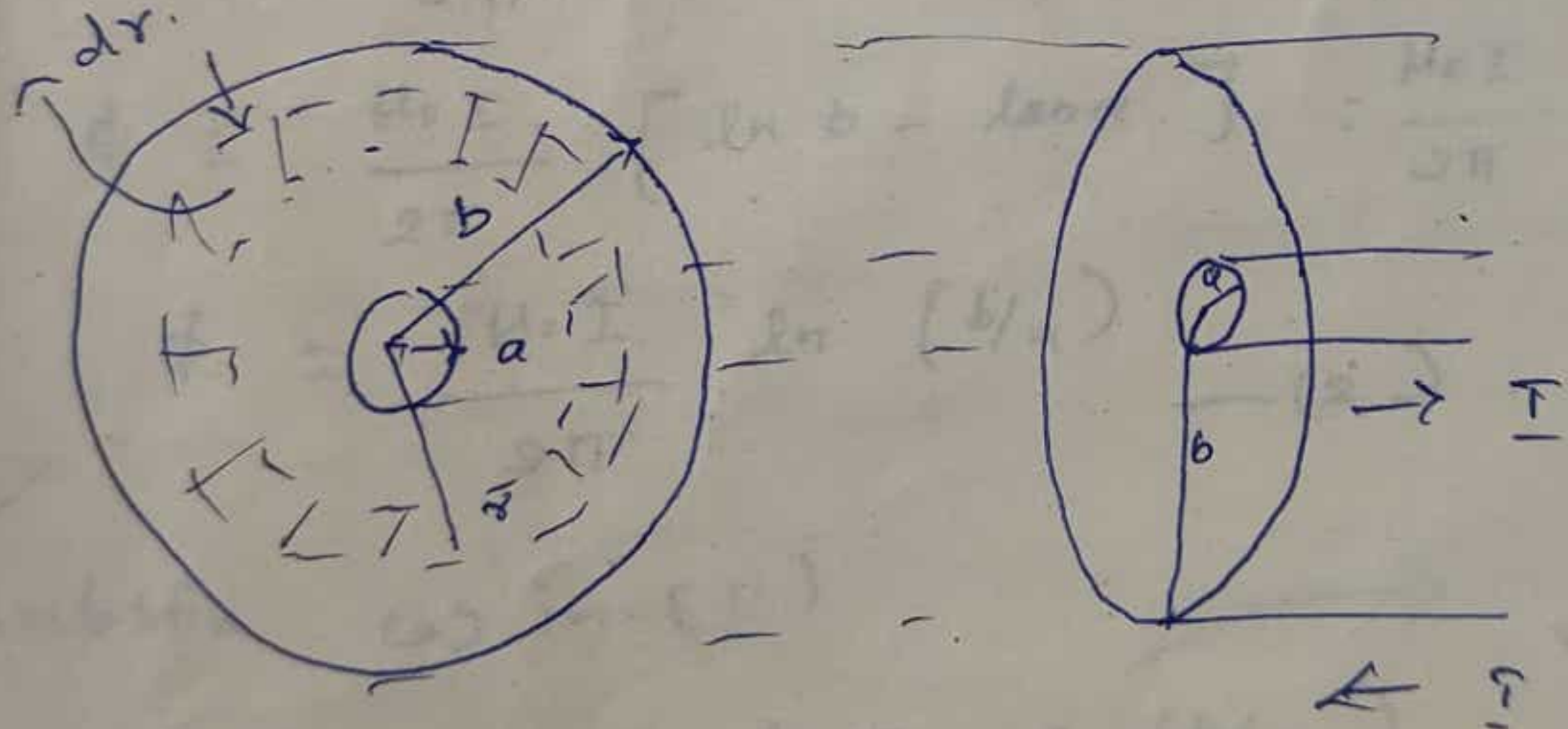
relation between M and L

$$M = K \sqrt{L_1 L_2}$$

$$K = \frac{M}{\sqrt{L_1 L_2}}$$

$$\therefore K < 1$$

Inductance of coaxial cable.



Inductance of coaxial cable:

self inductance is defined as.

$$L = \frac{N\Phi}{I}$$

For a coaxial cable, $N=1$ [only one circle].

$$L = \frac{\Phi}{I} \quad \text{--- (1)}$$

Laplace Equation:-

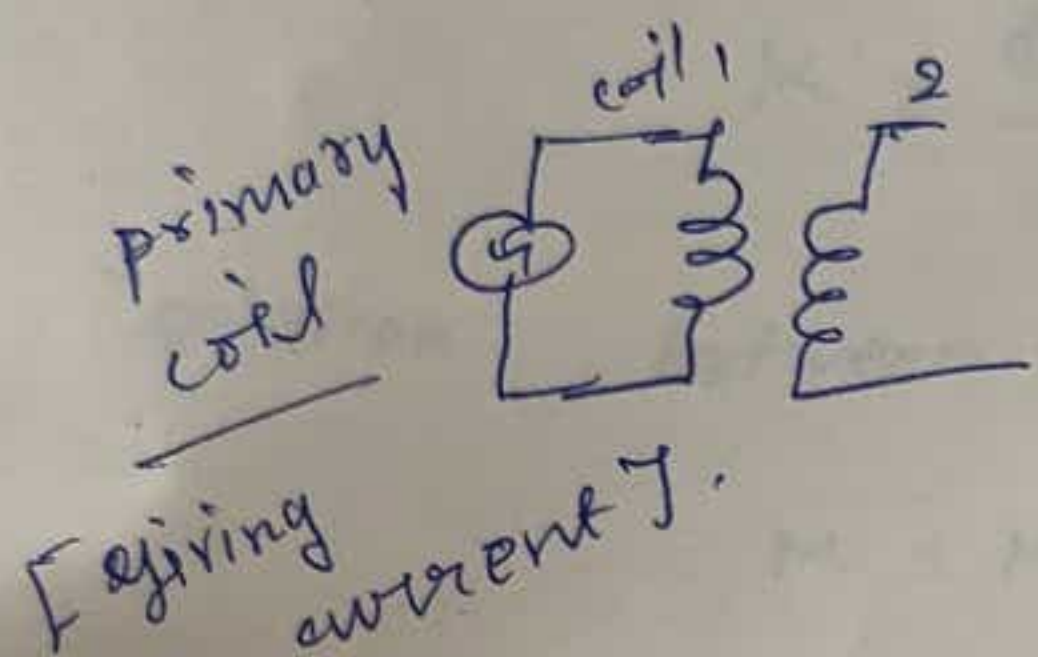
Main condition $\nabla^2 V = 0$

Mutual Inductance.

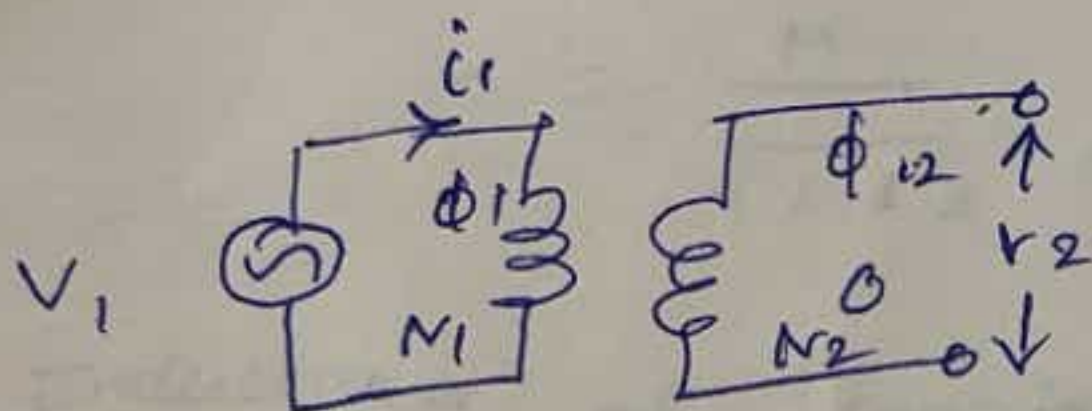
Def.

Mutual inductance between two coils is defined as the ratio of induced magnetic flux linkage in one coil to the current through in other coil.

$$M = \frac{N_1 \Phi_{21}}{i_2} ; \text{ Henry.}$$



Proof:



→ consider two coils magnetically coupled together.

→ Φ_{12} is produced by the current i_1

→ the induced emf V_2 in coil 2 is proportional to the rate of change of ' i_1 '

$$V_2 \propto \frac{di_1}{dt}$$

$$V_2 = M \frac{di_1}{dt} \quad \text{--- (1)}$$

From Faraday's law, the induced emf in coil 2 is.

$$V_2 = N_2 \frac{d\Phi_{12}}{dt} \quad \text{--- (2)} \quad \text{where } V \propto \frac{d\Phi}{dt}$$

Equating (1) & (2)

$$M \frac{di_1}{dt} = N_2 \frac{d\Phi_{12}}{dt}$$

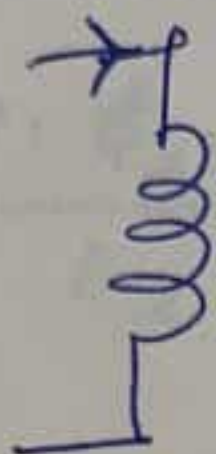
$$M = N_2 \frac{d\Phi_{12}}{di_1} \quad \text{--- (3)}$$

$$\frac{N_1 \Phi_{21}}{i_2}$$

Self Inductance: (L)

It is defined as the ratio of total magnetic flux linkage to the current through the coil.

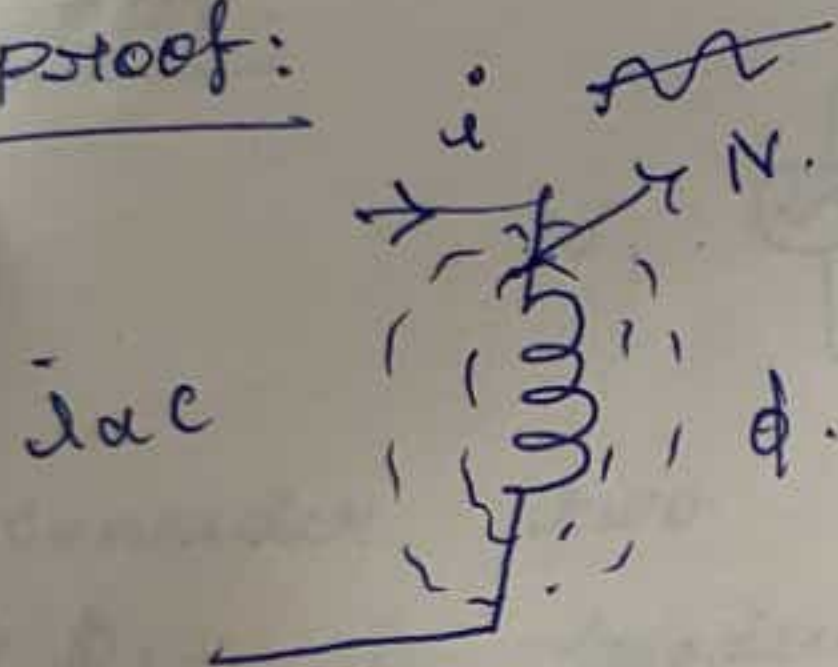
based on
i/p
current.



$$L = \frac{N\phi}{i} ; \text{ Henry.}$$

where; $N \rightarrow$ no. of turns.
 $\phi \rightarrow$ magnetic flux
 $i \rightarrow$ current.

Proof:



$\rightarrow$ consider a coil having 'N' Number of turns.

$\rightarrow$ If changing current (a.c) is applied, the emf is induced in the coil.

$\rightarrow$ The induced emf is proportional to the rate of change of current.

$$V \propto \frac{di}{dt} ; V = L \frac{di}{dt} \quad (1)$$

where $\rightarrow$

$L \rightarrow$ self inductance (proportionality constant).

From Faraday's law,

$$V = N \frac{d\phi}{dt} \quad (2)$$

Equating (1) & (2)

$$L \frac{di}{dt} = N \frac{d\phi}{dt}$$

$$L = N \frac{d\phi}{di}$$

$$L = \frac{N\phi}{i} ; \text{ Henry.}$$

21. Advanced communication lab Incharge (22-odd).
(21-even)

22. 2024 (even sem) symposium co-ordinator.

23. (2021-22) Result Analysis Incharge.

24. M.E - Modern communication Technique (22-odd).

25. Digital Electronic lab Incharge (22-even)

26. " " Theory. (22-even).

27. Wireless Network (22-even)

28. MPMC lab Incharge (21-odd).

29. Linear digital Integrated ckt (21-even).

30. Opto electronics (21-even).

31.

→ since the current is entering both the coils at the same terminal, it is called series aiding connection.

→ By applying kirchoff's law.

$$L_1 \frac{di}{dt} + M \frac{di}{dt} + L_2 \frac{di}{dt} + M \frac{di}{dt} = v(t)$$

$$(L_1 + L_2 + 2M) \frac{di}{dt} = v(t)$$

The equivalent inductance is,

$$L_{eq} = L_1 + L_2 + 2M$$

Henry.

Project Seminar
9th

1. BOS Member.
2. IA co-ordinator.
3. FA - (2024) March April.
4. AC (2025)
5. LIC (incharge - 2025) | (TECH Microwave lab incharge) - 2025.
lab
6. WC, FOC subject (24 odd).
7. TECH lab. (Incharge). - For 4 years.
8. Wireless Network (2024 - odd sem) (theory).
9. 23 (~~even~~ odd) wireless communication.
10. communication Electronic circuits (odd (23)).
11. project 1 Incharge co-ordinator (23 odd).
12. Mentor final year students (25 - final year).
13. Network circuit theory (23 - even) Boys.
14. Digital electronics lab (23 - even) Incharge.
15. value edu. (Audit course) ME - 23 even.
16. Automotive electronic system - 23 even.
17. Wireless communication (22 - odd)
18. " Network (22 - odd)
19. project work - 1 (22 - odd) (co-ordinator)
20. Minutes of Meeting (22 - odd) Incharge.

problem:

A coaxial cable has 2 conductors aligned with radius a of the core conductor and 1mm of the outer conductor.

If the insulator is in free space, determine the inductance of the coaxial cable:

sol:

$$L = \frac{\mu_0}{2\pi} \ln(b/a)$$

$$= \frac{2 \times 10^{-7} \times \ln(1/10^{-3})}{1}$$

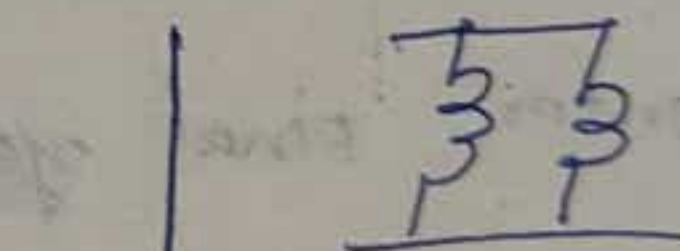
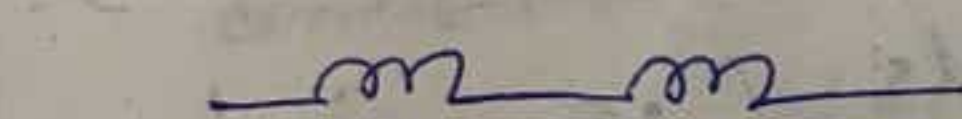
$$= 2 \times 10^{-7} \times \ln(10^3)$$

$\Rightarrow$ Inductance of coupled coils:

coupled coils

Series connection

Parallel connection

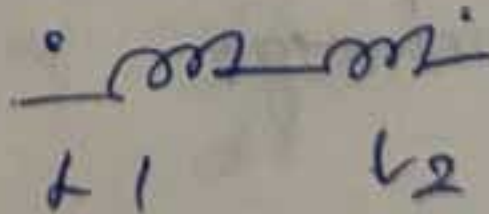
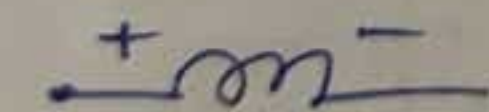


Aiding

opposing

Aiding

opposing



Aiding series connection:

